

Wealth Inequality and Financial Development: Revisiting the Symmetry Breaking Mechanism*

Haiping Zhang[†]

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Abstract

In an overlapping generations model with financial frictions and the fixed investment size requirement, Matsuyama (2004, *Econometrica*) shows that, in the absence of integrated financial markets, the world economy has a unique steady state, which is symmetric and stable in the sense that inherently identical countries converge to the same income level in the long run, regardless of their initial income level; financial globalization may “*break*” this *symmetric* steady state and lead to cross-country income polarization. He calls this phenomenon “symmetry breaking” and points out that financial underdevelopment is one of the necessary conditions.

We revisit this result by introducing wealth inequality and the minimum investment requirement into his framework. Increasing wealth inequality strictly reduces the possibility of symmetry breaking; if wealth inequality exceeds a threshold value, symmetry breaking does not arise at all, regardless of the level of financial development. Thus, wealth inequality is an equally important factor as financial development in determining the possibility of symmetry breaking.

We also address some practical issues in this framework, e.g., the conditions of financial integration, the domestic financial crisis and capital controls, and the world interest rate shocks and income volatility.

Keywords: financial frictions, financial globalization, minimum investment requirements, symmetry breaking, wealth inequality

JEL Classification: E44, F41

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[†]School of Economics, Singapore Management University. 90 Stamford Road, Singapore 178903.
E-mail: zhanghaiping2001@yahoo.com

1 Introduction

Financial globalization is one of the most fiercely debated topics of our times. Some economists argue that greater openness to financial flows has, by and large, proven essential for countries aiming to upgrade from lower to middle income status, while significantly enhancing stability among industrialized countries (Fischer, 1998; Summers, 2000), while others are more cautious and consider increasing capital account liberalization and free capital mobility as a potential threat to the stability of global financial system (Bhagwati, 1998; Rodrik, 1998; Stiglitz, 2002). In 1997, the International Monetary Fund (hereafter, IMF) planned to include capital account convertibility among its mandates. However, the 1998 Asian financial crisis forced the IMF to put it on hold. The world-wide economic turmoil in the aftermath of the 2008 global financial crisis further forced the IMF to reconsider its position on capital account liberalization. In November 2012, the IMF embraced a new “institutional view”, acknowledging the fact that capital flows are more beneficial and less risky if countries reach certain levels or “thresholds” of financial and institutional development, and that liberalization needs to be well planned, timed, and sequenced to ensure that its benefits outweigh the costs (IMF, 2012a,b).

In the literature, financial frictions help explain many puzzling phenomena in the era of financial globalization, e.g., financial crises, capital flows reversal, sudden stops, excess volatilities, the Lucas puzzle, the allocation puzzle, the global imbalances, and etc. (Gourinchas and Rey, 2014; Korinek and Mendoza, 2014; Mendoza, Quadrini, and Rios-Rull, 2009). Matsuyama (2004) synthesizes the two conflicting views on financial globalization in a stylized model with financial frictions and the fixed investment size requirement (hereafter, FIR). Under international financial autarky (hereafter, autarky), the world economy has a unique steady state, which is symmetric and stable in the sense that inherently identical countries converge to the same income level in the long run, regardless of their initial income level. Financial globalization may “break” this *symmetric* steady state and lead to cross-country income polarization. He calls this phenomenon “symmetry breaking” and it happens only if the level of financial development in the integrated countries is below a threshold value; otherwise, if the level of financial development in the integrated countries is above that threshold value, financial integration unambiguously leads to income convergence. Thus, in order to avoid symmetry breaking, the financial sector development can be considered as a “threshold” condition for financial integration, which is consistent with the IMF’s new “institutional view”. Besides, Matsuyama (2004) mentions in subsection 7.1 and 7.2 that symmetry breaking may also arise in the presence of wealth inequality and the minimum investment requirements (hereafter, MIR), while a complete characterization of multiple steady states is “*hopelessly complicated*”.

In the meantime, income and wealth inequality is an equally controversial topic as financial globalization and attracts enormous attention from both academia and public media. After declining in the first half of the 20th century, income and wealth inequality makes a comeback, which has profound implications to investment, growth, welfare, social and political stability, and etc. (Atkinson, Piketty, and Saez, 2011; Piketty, 2014; Quadrini and Rios-Rull, 2015; Saez and Zucman, 2014). Antunes and Cavalcanti (2013); Jaumotte, Lall, and Papageorgiou (2013); Mendoza, Quadrini, and Rios-Rull (2007) inves-

tigate how financial globalization may affect the wealth inequality and welfare. However, the impacts of wealth inequality on financial globalization are not well studied.

In this paper, we connect these two strands of literature and analyze how wealth inequality may fundamentally reshape the consequences of financial globalization in the framework of Matsuyama (2004). In particular, we provide a complete, analytical characterization of symmetry breaking in the presence of wealth inequality and the MIR.¹ Increasing wealth inequality strictly reduces the possibility of symmetry breaking; if wealth inequality exceeds a threshold value, symmetry breaking does not arise at all, regardless of the level of financial development. Thus, wealth inequality is an equally important factor as financial development in determining the possibility of symmetry breaking.

The intuition is as follows. In a neoclassical, frictionless model, the interest rate coincides with the rate of return to investment and both decrease in aggregate income. It is a result of the decreasing marginal product of capital (hereafter, MPK), which we call the neoclassical effect. Suppose that the world economy consist of a continuum of inherently identical countries. Under autarky, the interest rate is higher in the poor than in the rich countries, due to the neoclassical effect; under financial globalization, capital flows are “downhill” from the rich to the poor countries, leading to income convergence, and eventually all countries reach the same income level.

In a model with financial frictions, the aggregate credit demand is constrained, which pushes the interest rate below the marginal rate of return to investment². Suppose that the individual investment is indivisible and subject to the MIR. Only the agents who have the sufficiently high net wealth can meet the MIR and operate the productive project. We call them *entrepreneurs*. Other agents can only lend out their wealth and we call them *households*. Since the interest rate is below the rate of return to investment, entrepreneurs prefer to borrow to limit. The higher the aggregate income, the higher the individual net wealth, the larger (smaller) the mass of entrepreneurs (households), the larger (smaller) the aggregate credit demand (supply) on the extensive margin. This way, higher aggregate income tends to push up the interest rate through the extensive margin channel on the credit market. Under autarky, if this extensive-margin effect dominates the neoclassical effect, the interest rate is higher in the rich than in the poor countries; under financial globalization, capital flows are “uphill” from the poor to the rich countries so that countries are endogenously divided into two groups with different income levels.

Symmetry breaking is essentially a result of the positive interest rate response to income changes which ultimately depends on the size of the extensive-margin effect. Through the extensive-margin channel, financial development and wealth inequality affect the interest rate response and then the possibility of symmetry breaking.

- The lower the level of financial development, the tighter the borrowing constraints, the larger the wedge between the interest rate and the rate of return to investment.

Under certain conditions, it is more likely that the extensive-margin effect dominates

¹The FIR in Matsuyama (2004) $m_t = \mathbf{m}$ is a two-side restriction, i.e., the individual investment size can neither be lower nor greater than a specific value, while the MIR in our model $m_t \geq \mathbf{m}$ is a one-side restriction. Thus, we relax his assumption on the investment requirement. Besides, his model is virtually a special case of ours when the wealth distribution in our model is degenerated into a unit mass.

²It holds as long as the aggregate credit supply is not perfectly elastic to the interest rate.

the neoclassical effect and the interest rate responds positively to income changes. It explains the crucial role of financial development in Matsuyama (2004).

- The larger the wealth inequality, the less responsive the aggregate credit demand and supply to income changes, the weaker the extensive-margin effect, the more likely the neoclassical effect may dominate the extensive-margin effect and the interest rate responds negatively to income changes. It explains the crucial role of wealth inequality in our model.

Then, we use this model to address three practical issues.

First, whether a country should allow financial integration depends on its level of financial development relative to other financially integrated countries and possibly on its income level. It is consistent with the empirical evidence (Kose, Prasad, Rogoff, and Wei, 2009; Kose, Prasad, and Taylor, 2011). Kose, Prasad, Rogoff, and Wei (2010) show that financial integration leads to better macroeconomic outcomes when certain threshold conditions are met. Capital account liberalization in the absence of essential supporting conditions can vitiate the benefits, while economic policies designed to foster these necessary supporting conditions could be instrumental in more effectively utilizing the growth gains stemming from financial integration. Here, the extent of financial sector development is one of the threshold conditions.

Second, for a mild and temporary domestic financial crisis, a country does not need to impose capital controls, as the economy can recover to its pre-crisis level when the financial sector resumes its functions; for a severe and prolonged domestic financial crisis, the less financially developed country may impose capital controls to avoid the economy to be trapped in a low-income steady state. It is consistent with the IMF’s new “institutional view” that capital outflow management measures can be introduced temporarily in crisis situations or when a crisis may be imminent (IMF, 2015).

Third, in the case of the world interest rate shocks, income volatility is much higher in the less than in the more financially developed countries. Different from the traditional business cycles literature that analyzes income volatility with the log-linear approximation around the steady state, we analyze the whole state space of endogenous variables and the world interest rate shocks may bring a less financially developed country to a entirely different steady state with income level far from the previous one.

The MIR is a key assumption in our model. It has been extensively used in the literature to capture the investment indivisibility at the individual level, which is an important feature of business ideas, physical and human capital (Aghion and Bolton, 1997; Banerjee and Moll, 2010; Banerjee and Newman, 1993; Galor and Zeira, 1993; Piketty, 1997). In the recent literature (Barseghyan and DiCecio, 2011; Buera, Kaboski, and Shin, 2011; Erosa and Hidalgo-Cabrillana, 2008; Manova, 2013; Midrigan and Xu, 2014), the fixed costs or the entry costs are commonly introduced at the firm level and, in equilibrium, the individual investment is above a minimum scale. In other words, introducing either the MIR or the fixed cost makes the individual production set non-convex and, in the presence of financial frictions, a change in aggregate income affects the individual’s net wealth and the mass of investors so that aggregate investment adjusts on the extensive margin. The MIR in our model serves as a short-cut to feature this

mechanism. It significantly simplifies our analysis and allows us to characterize the dynamic properties for the entire parameter spaces analytically.

The rest of the paper is structured as follows. Section 2 describes the model with financial frictions and MIR. Section 3 analyzes the equilibrium allocation under autarky and shows the importance of the extensive-margin effect in generating the positive interest rate response to income changes. Section 4 explains intuitively the mechanism behind the symmetry-breaking result and highlights the role of wealth inequality. Section 5 addresses three practical issues. Section 6 discusses our future research agenda. Technical proofs and relevant discussions are included in the appendix.

2 The Model

The world economy consists of a continuum of countries, indexed by $i \in [0, 1]$. Countries are inherently identical except for the initial income level. In each country, a continuum of agents indexed by $j \in [0, 1]$ are born every period and live for two periods, young and old; agent j is endowed with $l_j = (1 - \theta)\epsilon_j$ units of labor when young and consumes only when old, where $\epsilon_j \in (1, \infty)$ follows the Pareto distribution with the cumulative distribution function $G(\epsilon_j) = 1 - \epsilon_j^{-\frac{1}{\theta}}$ and $\theta \in (0, 1)$.³ The population size of each generation is constant at one. In equilibrium, agents supply the labor endowment inelastically to the market and the aggregate labor supply is constant at $L = \int_1^\infty l_j dG(\epsilon_j) = 1$.

A final good is internationally tradable and chosen as the numeraire. The final good can be consumed or used to produce capital goods, which becomes available in the next period. Capital goods are non-tradable and can be combined with labor to produce final goods contemporaneously. Capital fully depreciates after the production. The markets for final goods, capital goods, and labor are perfectly competitive. There is no uncertainty in the model economy. Y_t^i denotes aggregate output of final goods, $L = 1$ and K_t^i denote the aggregate inputs of labor and capital goods, w_t^i and q_t^i denote the wage rate and the price of capital goods in country i and period t . To sum up,

$$Y_t^i = \left(\frac{K_t^i}{\alpha}\right)^\alpha \left(\frac{L}{1 - \alpha}\right)^{1 - \alpha}, \quad \text{where} \quad \alpha \in (0, 1), \quad (1)$$

$$q_t^i K_t^i = \alpha Y_t^i \quad \text{and} \quad w_t^i L = (1 - \alpha) Y_t^i. \quad (2)$$

Each agent is endowed with an indivisible project to produce capital goods subject to the MIR. Consider agent j born in country i and period t . As shown in the left panel of figure 1, the agent invest $m_{j,t}^i$ units of final goods in period t and produce $k_{j,t+1}^i = Rm_{j,t}^i$ units of capital goods in period $t + 1$, if its investment size is no less than a specific value,

³The inverse of θ is the tail index of the Pareto distribution. Pareto distribution has been widely used to feature the income and wealth distribution in the literature (Atkinson, Piketty, and Saez, 2011; Gabaix, 2009; Jones, 2015). In particular, the top tail of the income distribution is very well approximated by a Pareto distribution (Kuznets and Jenks, 1953; Piketty, 2014; Piketty and Saez, 2003). Besides the empirical relevance, assuming the Pareto distribution for the labor endowment also allows us to obtain the analytical solutions. Alternatively, one can also use other forms of distribution, e.g., the uniform distribution for labor endowment, e.g., $l_j \in [1 - a, 1 + a]$ with the probability density function $\frac{1}{2a}$. It makes the analysis much more complicated, without changing our qualitative results.

$m_{j,t}^i \geq m_t^i$; otherwise, the output is zero.⁴ The MIR has the function form $m_t^i = \mathbf{m}(Y_t^i)^{1-\sigma}$ with $\mathbf{m} > 0$. As shown in the right panel of figure 1, the MIR is constant at $\mathbf{m}$ for $\sigma = 1$ and linear in aggregate income for $\sigma = 0$. This function form allows for the possibility that the MIR may differ in the rich and in the poor country.⁵

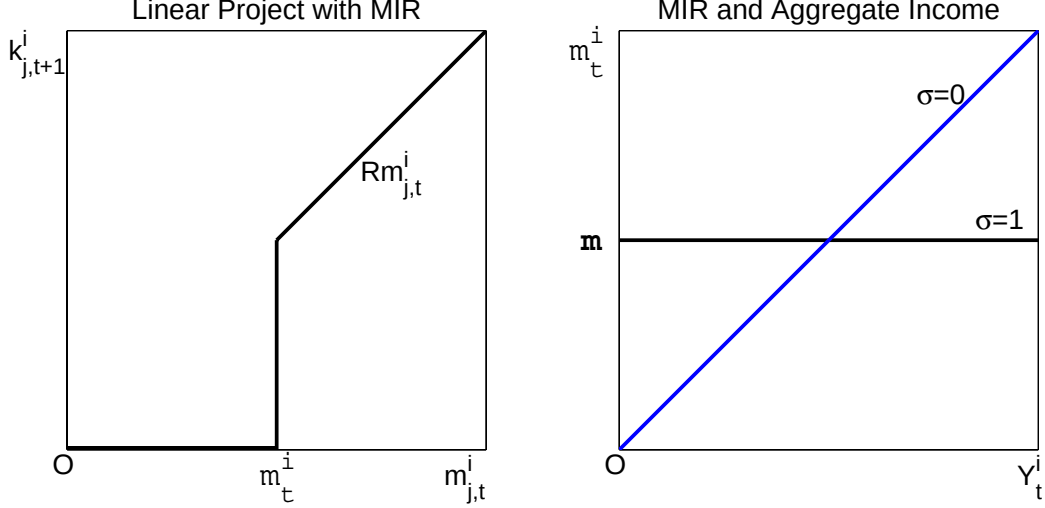


Figure 1: Individual Investment Project, MIR, and Aggregate Income

Agents can save the labor income $n_{j,t}^i = w_t^i l_j$ either by producing capital goods at the marginal rate of return $q_{t+1}^i R$ or lending out in the credit market at the gross interest rate r_t^i . The interest rate cannot exceed the marginal rate of return $r_t^i \leq q_{t+1}^i R$; otherwise, nobody would produce capital goods.

Let us start with the case of $r_t^i < q_{t+1}^i R$. If agent j can meet the MIR, it prefers to finance its investment, $m_{j,t}^i$, with loans. However, due to limited commitment, it can only borrow up to a fraction λ of its investment return in the present value and has to use its own funds as equity capital to cover the gap,

$$b_{j,t}^i \leq \lambda \frac{q_{t+1}^i R m_{j,t}^i}{r_t^i}, \quad \text{and} \quad m_{j,t}^i - b_{j,t}^i \leq n_{j,t}^i, \quad (3)$$

where $\lambda \in (0, 1)$ reflects the level of financial development.⁶ Let $\psi_{j,t}^i \equiv \frac{m_{j,t}^i - b_{j,t}^i}{m_{j,t}^i}$ denote the agent's equity-investment ratio. In period $t + 1$, it gets the investment return, $q_{t+1}^i R m_{j,t}^i$, repays the debt, $r_t^i b_{j,t}^i$, and consumes the rest. The equity rate is defined as the rate of return to its equity capital, $\Omega_{j,t}^i \equiv \frac{q_{t+1}^i R m_{j,t}^i - r_t^i b_{j,t}^i}{m_{j,t}^i - b_{j,t}^i}$. Use the borrowing constraint (3) to get,

⁴Despite the nonconvex individual production set, Matsuyama (2007, 2008) argues that assuming a continuum of agents convexifies the aggregate production set.

⁵We assume the dependence of the MIR on aggregate income mainly for the analytical purpose. As shown in section 3, for $\sigma = 0$, capital accumulation does not affect the mass of investors so that the aggregate investment responds only on the intensive margin; for $\sigma = 1$, capital accumulation affects the mass of investors so that aggregate investment responds on the intensive and extensive margins. We highlight the role of the extensive-margin channel by comparing the results in the two cases.

⁶Matsuyama (2008) shows that the strategic default à la Hart and Moore (1994) can give rise to this form of the borrowing constraints.

$$\psi_{j,t}^i \geq 1 - \lambda \frac{q_{t+1}^i R}{r_t^i}, \quad (4)$$

$$\Omega_{j,t}^i = \underbrace{q_{t+1}^i R}_{\text{the project's rate of return}} + \underbrace{(q_{t+1}^i R - r_t^i) \left(\frac{1}{\psi_{j,t}^i} - 1 \right)}_{\text{the leverage effect}}. \quad (5)$$

The equity rate exceeds the project's rate of return $q_{t+1}^i R$ by the amount of $(q_{t+1}^i R - r_t^i) \left(\frac{1}{\psi_{j,t}^i} - 1 \right)$, which is called the leverage effect and depends positively on the spread $(q_{t+1}^i R - r_t^i)$ and the debt-equity ratio $\left(\frac{1}{\psi_{j,t}^i} - 1 \right)$. Given the positive spread $q_{t+1}^i R > r_t^i$, the agent maximizes the leverage effect by borrowing to the limit so that equality holds in (4) and $\psi_{j,t}^i$ does not depend on the agent's net wealth. The positive leverage effect induces the agent to invest its entire labor income in the project, $m_{j,t}^i - b_{j,t}^i = n_{j,t}^i$.

If $r_t^i = q_{t+1}^i R$, the leverage effect vanishes and $\Omega_{j,t}^i = r_t^i$. Thus, the agent is indifferent between investing in the project or lending. In this case, $m_{j,t}^i$ and $\psi_{j,t}^i$ are indeterminate. To sum up,

$$\psi_{j,t}^i \begin{cases} = \psi_t^i \equiv 1 - \lambda \frac{q_{t+1}^i R}{r_t^i}, & \text{and is wealth-independent,} & \text{if } r_t^i < q_{t+1}^i R; \\ \geq 1 - \lambda \frac{q_{t+1}^i R}{r_t^i}, & \text{and is indeterminate,} & \text{if } r_t^i = q_{t+1}^i R; \end{cases} \quad (6)$$

$$\Omega_{j,t}^i = \Omega_t^i = \begin{cases} q_{t+1}^i R + (q_{t+1}^i R - r_t^i) \left(\frac{1}{\psi_t^i} - 1 \right) > q_{t+1}^i R, & \text{if } r_t^i < q_{t+1}^i R; \\ q_{t+1}^i R, & \text{if } r_t^i = q_{t+1}^i R; \end{cases} \quad (7)$$

$$m_{j,t}^i \begin{cases} = \frac{n_{j,t}^i}{\psi_t^i} = \frac{w_t^i}{\psi_t^i} (1 - \theta) \epsilon_j, & \text{and is linear in labor income,} & \text{if } r_t^i < q_{t+1}^i R; \\ \leq \frac{n_{j,t}^i}{\psi_t^i}, & \text{and is indeterminate,} & \text{if } r_t^i = q_{t+1}^i R. \end{cases} \quad (8)$$

If $r_t^i < q_{t+1}^i R$, there exists a cutoff value $\underline{\epsilon}_t^i$. The agents with $\epsilon_j \geq \underline{\epsilon}_t^i$ can meet the MIR, $m_{j,t}^i = \frac{w_t^i}{\psi_t^i} (1 - \theta) \epsilon_j \geq m_t^i$, and are called *entrepreneurs*. Their total mass is $\tau_t^i = (\underline{\epsilon}_t^i)^{-\frac{1}{\theta}}$. $\underline{\epsilon}_t^i$ is associated with the entrepreneur who just meets the MIR,

$$m_{j,t}^i(\underline{\epsilon}_t^i) = \frac{w_t^i}{\psi_t^i} (1 - \theta) \underline{\epsilon}_t^i = \mathbf{m} (Y_t^i)^{1-\sigma}, \Rightarrow \underline{\epsilon}_t^i = \frac{\psi_t^i \mathbb{F}}{(w_t^i)^\sigma}, \text{ where } \mathbb{F} \equiv \frac{\mathbf{m}}{(1 - \alpha)^{1-\sigma} (1 - \theta)}.$$

The constant size of population in each generation imposes a natural constraint on the mass of entrepreneurs, i.e., $\tau_t^i \leq 1$. Thus, the cutoff value is

$$\underline{\epsilon}_t^i = \max \left\{ 1, \frac{\psi_t^i \mathbb{F}}{(w_t^i)^\sigma} \right\}. \quad (9)$$

When young, entrepreneurs use the entire labor income, $n_{j,t}^i$, and the loan $b_{j,t}^i = n_{j,t}^i \left(\frac{1}{\psi_t^i} - 1 \right)$ to finance their investment; when old, they consume, $c_{j,t+1}^{i,e}$, and exit from the economy,

$$n_{j,t}^i = w_t^i l_j \quad \text{and} \quad c_{j,t+1}^{i,e} = n_{j,t}^i \Omega_t^i. \quad (10)$$

The agents with $\epsilon_j \in [1, \underline{\epsilon}_t^i)$ cannot meet the MIR and they are called *households*. Their total mass is $1 - \tau_t^i = 1 - (\underline{\epsilon}_t^i)^{-\frac{1}{\theta}}$. When young, households lend out their entire labor income $n_{j,t}^i$; when old, they consume, $c_{j,t+1}^{i,h}$, and exit from the economy,

$$n_{j,t}^i = w_t^i l_j \quad \text{and} \quad c_{j,t+1}^{i,h} = n_{j,t}^i r_t^i. \quad (11)$$

We analyze the equilibrium allocation under two scenarios: (1) autarky where agents are allowed to borrow or lend domestically, (2) financial globalization where agents are allowed to borrow or lend domestically as well as internationally.⁷

Let M_t^i , D_t^i , and S_t^i denote the aggregate investment, the aggregate credit demand and supply. Under autarky, the markets for capital goods and credit clear domestically,⁸

$$K_{t+1}^i = \int_{\underline{\epsilon}_t^i}^{\infty} Rm_{j,t}^i dG(\epsilon_j) = RM_t^i, \text{ where } M_t^i \equiv \int_{\underline{\epsilon}_t^i}^{\infty} m_{j,t}^i dG(\epsilon_j), \quad (12)$$

$$D_t^i \equiv \int_{\underline{\epsilon}_t^i}^{\infty} (m_{j,t}^i - n_{j,t}^i) dG(\epsilon_j), \quad S_t^i \equiv \int_1^{\underline{\epsilon}_t^i} n_{j,t}^i dG(\epsilon_j), \quad D_t^i = S_t^i, \Rightarrow M_t^i = w_t^i. \quad (13)$$

If $r_t^i = q_{t+1}^i R$, those who can meet the MIR may not invest their entire labor income or borrow to the limit. Despite the indeterminacy at the individual level, aggregate saving is still fully used in the production of capital goods in equilibrium, $K_{t+1}^i = RM_t^i = R w_t^i$.

Definition 1. *Under autarky, a market equilibrium in country i is a set of choices of agents, $\{n_{j,t}^i, m_{j,t}^i, c_{j,t}^{i,e}, c_{j,t}^{i,h}, \psi_{j,t}^i\}$, the threshold value $\{\underline{\epsilon}_t^i\}$, the prices $\{q_t^i, w_t^i, \Omega_t^i, r_t^i\}$, and the aggregate quantities $\{Y_t^i, K_t^i, M_t^i\}$, satisfying equations (1)-(2), (6)-(13).*

Under financial globalization, let ϕ_t^i denote the ratio of capital outflows over domestic saving in country i , with negative values indicating the case of capital inflows. The equilibrium conditions are identical as under autarky except for the domestic and world credit market conditions,

$$M_t^i = w_t^i(1 - \phi_t^i), \quad (14)$$

$$\int_0^1 w_t^i \phi_t^i di = 0. \quad (15)$$

Definition 2. *Under financial globalization, a market equilibrium in country i is a set of choices of agents, $\{n_{j,t}^i, m_{j,t}^i, c_{j,t}^{i,e}, c_{j,t}^{i,h}, \psi_{j,t}^i\}$, the threshold value $\{\underline{\epsilon}_t^i\}$, the prices $\{q_t^i, w_t^i, \Omega_t^i\}$, and the aggregate quantities $\{Y_t^i, K_t^i, M_t^i, \phi_t^i\}$, satisfying equations (1)-(2), (6)-(12), (14), the interest rate is equalized across countries $r_t^i = r_t^*$ and determined by equation (15).*

3 Equilibrium under Autarky

Without loss of generality, we suppress the country index i for the scenario of autarky. Let X_A denote the steady-state value of variable X_t under autarky. Given the fixed aggregate labor input $L = 1$, equation (2) implies that $w_t = (1 - \alpha)Y_t$. Thus, we use the wage rate as a proxy for aggregate income in the following analysis.

Combine (12)-(13) to get $K_{t+1} = R w_t$. Then, the law of motion for the wage rate⁹ is

$$w_{t+1} = \frac{(1 - \alpha)}{L} Y_{t+1} = \left(\frac{R w_t}{\rho} \right)^\alpha, \text{ where } \rho \equiv \frac{\alpha}{1 - \alpha}. \quad (16)$$

⁷Following Matsuyama (2004), we exclude FDI flows by assumption. von Hagen and Zhang (2014a,b) analyze the joint determination of financial capital flows and FDI flows in the setting without the MIR.

⁸According to the Walras' law, the market for final goods clears in equilibrium.

⁹Using the law of motion for wage simplifies our analysis. One can also use the law of motion for capital, which, however, complicates the analysis.

Proposition 1. *Under autarky, there exists a unique, stable steady state in each country where the wage rate is $w_A = \left(\frac{R}{\rho}\right)^\rho$.*

Rewrite equation (16) into $\ln w_{t+1} - \ln w_t = (\alpha - 1)(\ln w_t - \ln w_A)$. The decreasing MPK ($\alpha < 1$) acts as the convergence force that brings country i towards the steady state. Although the MIR, financial frictions, and wealth inequality do not affect the dynamics and the steady state under autarky in this model, they affect the interest rate response to income changes, which may lead to multiple steady states under financial integration.

Interest Rate Response to Income Changes

Iff $r_t < q_{t+1}R$, the borrowing constraints are binding for all entrepreneurs and the interest rate is a function of w_t defined jointly by equations (17)-(18).¹⁰

$$r_t = \frac{\lambda}{1 - \psi_t} q_{t+1} R = \frac{\lambda}{1 - \psi_t} w_t^{-(1-\alpha)} R^\alpha \rho^{1-\alpha}, \quad (17)$$

$$\psi_t = \left(\frac{w_t^\sigma}{\mathbb{F}} \right)^{1-\theta}. \quad (18)$$

Iff $r_t = q_{t+1}R$, the borrowing constraints are slack and the interest rate is,

$$r_t = q_{t+1}R = w_t^{-(1-\alpha)} R^\alpha \rho^{1-\alpha}. \quad (19)$$

As mentioned in section 2, the zero spread $r_t = q_{t+1}R$ leads to the indeterminacy of $m_{j,t}$ and $\psi_{j,t}$. For analytical simplicity, we focus on an equilibrium where all entrepreneurs still invest their entire labor income and choose the same ψ_t determined by equation (18).

Define $\Lambda \equiv \frac{(1-\alpha)(1-\theta)}{(1-\lambda)^{\frac{1}{1-\theta}}}$ as an increasing function of $\lambda \in (0, 1)$. Let $\bar{\psi} \equiv 1 - \lambda$.

Lemma 1. *Iff $\mathbf{m} \leq Y_t^\sigma \Lambda$, the borrowing constraints are slack and $\psi_t \in (\bar{\psi}, 1)$; iff $\mathbf{m} \geq Y_t^\sigma \Lambda$, the borrowing constraints are binding and $\psi_t \in (0, \bar{\psi}]$.*

If the borrowing constraints are slack, the interest rate declines in aggregate income, due to the decreasing MPK. We call it the neoclassical effect. Use equation (19) to get

$$\frac{\partial \ln r_t}{\partial \ln w_t} = \underbrace{\frac{\partial \ln q_{t+1} R}{\partial \ln w_t}}_{\text{the neoclassical effect}} = \alpha - 1 < 0. \quad (20)$$

If the borrowing constraints are binding, the interest rate may rise in aggregate income. We derive the relevant conditions by analyzing the credit market equilibrium.

Use equations (3) and (13) to rewrite the aggregate credit demand and supply as $D_t = \frac{\lambda q_{t+1} R}{r_t} M_t = \frac{\lambda q_{t+1} R}{r_t} w_t$ and $S_t = w_t [1 - \underline{\epsilon}_t^{-(\frac{1}{\theta}-1)}]$, which are affected by various factors,

$$\ln D_t = \underbrace{\ln w_t}_{\text{net-wealth effect}} + \underbrace{\ln q_{t+1} R}_{\text{neoclassical effect}} + \underbrace{\ln \lambda}_{\text{financial-development effect}} - \underbrace{\ln r_t}_{\text{interest-rate effect}} \quad (21)$$

$$\ln S_t = \underbrace{\ln w_t}_{\text{net-wealth effect}} + \underbrace{\ln(1 - \tau_t^{1-\theta})}_{\text{supply-side extensive-margin effect}}. \quad (22)$$

¹⁰See the proof of lemma 1 in appendix C for derivation.

According to equation (21), the present value of the entrepreneurs' pledgeable investment return declines in the interest rate and hence, the credit demand curve is downward sloping, $\frac{\partial D_t}{\partial r_t} < 0$; according to equation (22), since households supply their entire labor income inelastically to the credit market, the credit supply curve is vertical $\frac{\partial S_t}{\partial r_t} = 0$. Besides, the aggregate credit demand and supply are affected by the following factors.

- The net-wealth effect: the higher the aggregate income, the higher the agents' labor income and net wealth, the higher the aggregate credit demand and supply.
- The neoclassical effect: the higher the aggregate investment in period t , the lower the MPK in period $t + 1$, the lower the unit pledgeable value of entrepreneurs' investment in period t , the lower the aggregate credit demand.
- The financial-development effect: the higher the level of financial development, the more funds entrepreneurs can borrow, the higher the aggregate credit demand.
- The supply-side extensive-margin effect: the larger the mass of households $1 - \tau_t$, the higher the aggregate credit supply.

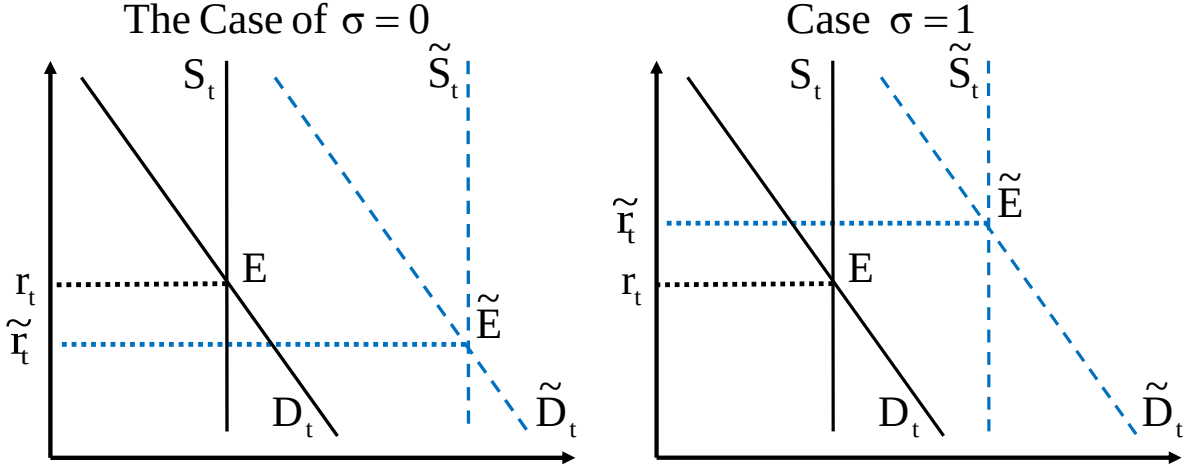


Figure 2: Credit Market Response to An Increase in Aggregate Income

Figure 2 shows the credit market equilibrium under autarky. The downward-sloping credit demand curve D_t and the vertical credit supply curve S_t intersect at point E with the equilibrium interest rate r_t . If aggregate income rises marginally from Y_t to $\tilde{Y}_t$, the aggregate saving rises proportionally from $w_t = (1 - \alpha)Y_t$ to $\tilde{w}_t = (1 - \alpha)\tilde{Y}_t$.

Combine equations (9) and (13) to get

$$\tau_t = \underline{\epsilon}_t^{-\frac{1}{\theta}} = \frac{w_t^\sigma}{\mathbb{F}}, \quad \text{and} \quad \frac{\partial \ln \tau_t}{\partial \ln w_t} = \sigma. \quad (23)$$

For $\sigma = 0$, a rise in Y_t raises the MIR $m_t = \mathbf{m}Y_t$ and the net wealth of individual agents $n_{j,t} = l_j w_t = l_j(1 - \alpha)Y_t$ in the equal proportions. According to equation (23), the cutoff value is constant at $\underline{\epsilon}_t = \underline{\epsilon}_A \equiv \mathbb{F}^\theta$ and so is the mass of entrepreneurs $\tau_t = \tau_A \equiv \frac{1}{\mathbb{F}}$. Thus, the aggregate credit demand and supply respond only on **the intensive margin**. According to equations (21) and (22), the net-wealth effect raises the credit supply and

demand in the equal proportions, while the neoclassical effect reduces the credit demand. As shown in the left panel of figure 2, the rightward shift of the credit demand curve is dominated by that of the credit supply curve and the credit market equilibrium moves from point E to $\tilde{E}$ with a lower interest rate $\tilde{r}_t < r_t$. In this case, the interest rate is purely driven by the neoclassical effect,

$$\begin{aligned} \frac{\partial \ln D_t}{\partial \ln w_t} &= 1 + \frac{\partial \ln q_{t+1} R}{\partial \ln w_t} - \frac{\partial \ln r_t}{\partial \ln w_t}, \text{ and } \frac{\partial \ln S_t}{\partial \ln w_t} = 1, \\ \frac{\partial \ln D_t}{\partial \ln w_t} &= \frac{\partial \ln S_t}{\partial \ln w_t}, \Rightarrow \frac{\partial \ln r_t}{\partial \ln w_t} = \underbrace{\frac{\partial \ln q_{t+1} R}{\partial \ln w_t}}_{\text{the neoclassical effect}} = \alpha - 1 < 0. \end{aligned} \quad (24)$$

For $\sigma = 1$, the MIR is constant at $m_t = \mathbf{m}$ and a rise in Y_t raises the net wealth of individual agents so that more agents can meet the MIR and become entrepreneurs, i.e., $\frac{\partial \ln \tau_t}{\partial \ln w_t} = \sigma > 0$, according to equation (23). Thus, the aggregate credit demand and supply respond on **the intensive and the extensive margins**. In particular, the decline in the mass of households reduces the credit supply on the extensive margin, which tends to raise the interest rate. Use equations (21)-(23) to get

$$\frac{\partial \ln r_t}{\partial \ln w_t} = \underbrace{\frac{\partial \ln q_{t+1} R}{\partial \ln w_t}}_{\text{the neoclassical effect}} - \underbrace{\frac{\partial \ln(1 - \tau_t^{1-\theta})}{\partial \ln \tau_t} \frac{\partial \ln \tau_t}{\partial \ln w_t}}_{\text{the extensive-margin effect}} = (\alpha - 1) + \frac{1 - \theta}{\frac{1}{\psi_t} - 1}. \quad (25)$$

If the extensive-margin effect dominates the neoclassical effect, the rightward shift of the credit supply curve is dominated by that of the credit demand curve and, as shown in the right panel of figure 2, the credit market equilibrium moves from point E to $\tilde{E}$ with a higher interest rate, $\tilde{r}_t > r_t$. Let $\tilde{\psi}_A \equiv \frac{1-\alpha}{1-\alpha+1-\theta}$ and $\tilde{\lambda}_A \equiv 1 - \tilde{\psi}_A$.

Lemma 2. *For $\sigma = 0$, the interest rate declines in aggregate income under autarky.*

For $\sigma = 1$, the interest rate rises in aggregate income under autarky, iff $\lambda \in (0, \tilde{\lambda}_A)$ and $w_t \in (\mathbb{F} \tilde{\psi}_A^{\frac{1}{1-\theta}}, \mathbb{F} \tilde{\psi}^{\frac{1}{1-\theta}})$ or equivalently $\psi_t = (\frac{w_t}{\mathbb{F}})^{1-\theta} \in (\tilde{\psi}_A, \tilde{\psi})$.¹¹

Given $\sigma = 1$, the left panel of figure 3 shows lemma 2 in the $\{\lambda, \psi_t\}$ space. In equilibrium, $\psi_t = (\frac{w_t}{\mathbb{F}})^{1-\theta} = \left[\frac{K_t^\alpha}{(L\rho)^\alpha \mathbb{F}} \right]^{1-\theta}$ and one can convert the threshold values from the $\{\lambda, \psi_t\}$ space to the $\{\lambda, K_t\}$ space, as shown in the right panel. For (λ, ψ_t) in region SD, the borrowing constraints are **slack** and, due to the neoclassical effect, the interest rate **declines** in aggregate income; for (λ, ψ_t) in region BI (BD), the borrowing constraints are **binding** and the interest rate **increases** (**declines**) in aggregate income, because the extensive-margin effect dominates (is dominated by) the neoclassical effect.

Figure 4 shows proposition 1 and lemma 2 graphically. In the left panel, the law of motion for wage under autarky crosses the 45° line once and only once from the left at point S with the wage rate w_A . In the middle panel, for $\sigma = 0$, the interest rate is

¹¹As long as the aggregate production function has the decreasing MPK, i.e., $f'(k) > 0$ and $f''(k) < 0$, where $k \equiv \frac{K}{L}$, the neoclassical effect and the extensive-margin effect exist in the presence of the constant MIR and the binding borrowing constraints. Thus, our assumption of the Cobb-Douglas production function is not essential for the positive interest rate response to income changes.

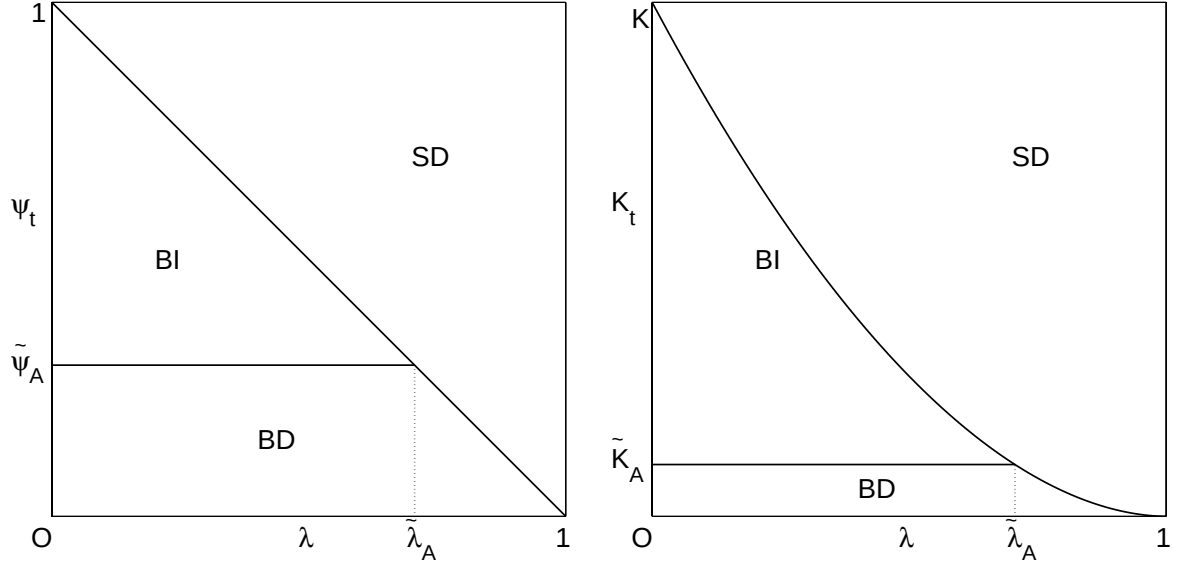


Figure 3: Interest Rate Response to Income Changes: $\sigma = 1$

proportional to the marginal rate of return to investment and, due to the neoclassical effect, they decline in w_t . In the right panel, for $\sigma = 1$ and $\lambda \in (0, \tilde{\lambda}_A)$, the extensive-margin effect interacts with the neoclassical effect so that the interest rate is a non-monotonic function of w_t .

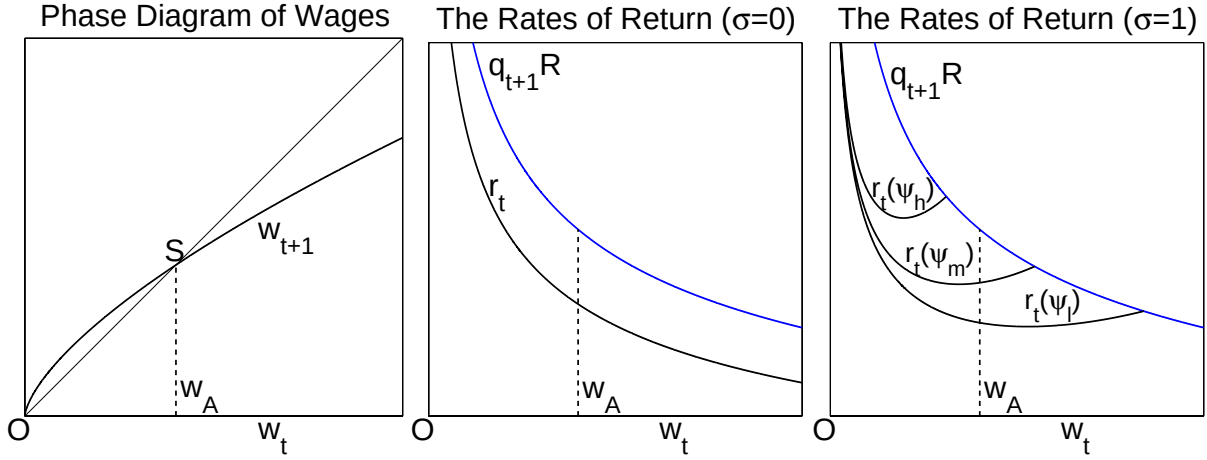


Figure 4: Wage Dynamics and Interest Rate Pattern

Combine equations (16) and (18) to get,

$$\psi_A = \left[\frac{w_A}{\mathfrak{m}} (1 - \theta) \right]^{1-\theta} = \left[\left(\frac{R}{\rho} \right)^\rho \frac{(1 - \theta)}{\mathfrak{m}} \right]^{1-\theta}. \quad (26)$$

Thus, the steady-state value ψ_A is a function of the model parameters $\{R, \mathfrak{m}, \rho, \theta\}$. If the parameter values are chosen in such a way that $\psi_A < \bar{\psi}$, the borrowing constraints are binding at the steady state. Evaluate equation (25) at the steady state,

$$\frac{\partial \ln r_t}{\partial \ln w_t} \Big|_{w_t=w_A} = \underbrace{\alpha - 1}_{\text{the neoclassical effect (-)}} + \underbrace{\frac{(1 - \theta)}{\frac{1}{\psi_A} - 1}}_{\text{the extensive-margin effect (+)}}. \quad (27)$$

Five factors affect the interest rate response to income changes around the steady state.

- The higher the α , the weaker the neoclassical effect.
- The lower the λ , the more likely the borrowing constraints are binding at the autarkic steady state ($\psi_A < 1 - \lambda$) and the extensive-margin effect in equation (27) is active.¹² Given $\psi_A < 1 - \lambda$,
 - The lower the θ , the smaller the within-country wealth inequality, the stronger the extensive-margin effect.
 - The higher the productivity measured by R or the lower the MIR measured by $\mathbf{m}$, the larger the mass of entrepreneurs and the ψ_A , the smaller the mass of households, the stronger the extensive-margin effect.

If $\psi_A = \psi_h \in (\bar{\psi}, 1)$ or $\psi_A = \psi_l \in (0, \tilde{\psi}_A)$, i.e., in region SD or BD of figure 3, the interest rate declines in w_t around the steady state; if $\psi_A = \psi_m \in (\tilde{\psi}_A, \bar{\psi})$, i.e., in region BI, the interest rate rises in w_t around the steady state. See the right panel of figure 4.

4 Equilibrium under Financial Globalization

We use the term “financial integration” from an individual country’s perspective and refer to the case where country i is integrated into the world financial market and its domestic interest rate is then equal to the world level. We use the term “financial globalization” from the world economy’s perspective and refer to the case where all countries are financially integrated and the world interest rate is endogenously determined.

Suppose that country i allows financial integration from period $t = 0$ on, given the world interest rate $r_t^i = r^*$. Without loss of generality, we assume $r^* = r_A$, where $r_A = \frac{\lambda}{1-\psi_A}\rho$ ($r_A = \rho$) if the borrowing constraints are binding (slack) in the autarkic steady state. Under financial globalization, the autarkic steady state is still a steady state, but it may not be stable or unique.

There exists a threshold value $\bar{w}_F$ such that for $w_t^i > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1}^i = r^*$, so that the law of motion for wage is flat at $w_{t+1}^i = w_A \left(\frac{\rho}{r^*}\right)^\rho$; for $w_t^i < \bar{w}_F$, the borrowing constraints are binding, $\psi_t^i < 1 - \lambda$, and

$$r_t^i = \frac{\lambda}{1-\psi_t^i} Rq_{t+1}^i = r^*, \quad \Rightarrow \quad w_{t+1}^i = w_A \left[\frac{\lambda}{(1-\psi_t^i)r^*} \rho \right]^\rho. \quad (28)$$

Under autarky, domestic investment is fully financed by domestic saving, $\frac{K_{t+1}^i}{R} = w_t^i$ and equation (16) specifies the law of motion for wage, regardless of whether the mass-of-entrepreneurs constraint is binding or not. Under financial integration, domestic investment depends on domestic saving and financial capital flows, $\frac{K_{t+1}^i}{R} = w_t^i(1 - \phi_t^i)$; whether

¹²If $\psi_A > 1 - \lambda$, the borrowing constraints are slack at the autarkic steady state where the extensive-margin effect is absent and the interest rate strictly declines in aggregate income.

the mass-of-entrepreneurs constraint is binding or not matters for the patterns of capital flows and domestic investment, which affects the law of motion for wage.¹³

- If the mass-of-entrepreneurs constraint is slack, $\tau_t < 1$, the law of motion for wage in country i are characterized by equations (28) and (29),

$$w_t^i = (\psi_t^i \mathbb{F}^{1-\theta})^{\frac{1}{\sigma(1-\theta)+\theta}} \left\{ \left[\frac{\lambda \rho}{r^*(1-\psi_t^i)} \right]^{\frac{1}{1-\alpha}} w_A \right\}^{\frac{\theta}{\sigma(1-\theta)+\theta}}; \quad (29)$$

- if the mass-of-entrepreneurs constraint is binding, $\tau_t = 1$, the law of motion for wage in country i are characterized by equations (28) and (30),

$$w_t^i = \psi_t^i \left[\frac{\lambda \rho}{r^*(1-\psi_t^i)} \right]^{\frac{1}{1-\alpha}} w_A. \quad (30)$$

As shown below, multiple steady states may arise in country i under financial integration, if the response of the autarkic interest rate to income changes is positive and strong enough. It depends on the form of the MIR, the dispersion of within-country wealth distribution, the borrowing constraints, and the mass-of-entrepreneurs constraint.

4.1 Dynamic and Steady-State Analysis: the Case of $\sigma = 0$

According to lemma 2, in the case of $\sigma = 0$, the autarkic interest rate declines in aggregate income, due to the neoclassical effect. Given $r^* = r_A$, if country i is initially above the autarkic steady state $Y_0^i > Y_A$, its autarkic interest rate is lower than the world level, $r_0^i < r^*$. From period $t = 0$ on, country i witnesses financial capital outflows, which makes its current domestic investment and next-period aggregate income lower than under autarky. The opposite applies to the case of $Y_0^i < Y_A$. The solid (dashed) curve in figure 5 shows the law of motion for wage under financial integration (autarky). The left (right) panel shows the case where the borrowing constraints are binding (slack) in the steady state.¹⁴ Due to the investment-dampening effect, the law of motion for wage around the autarkic steady state is flatter under financial integration than under autarky.

Proposition 2. *In the case of $\sigma = 0$, the autarkic steady state is still the unique, stable steady state under financial integration.*

Under financial integration, besides the decreasing MPK, the investment-dampening effect acts as another convergence force that brings country i towards the autarkic steady state. The investment-dampening effect results ultimately from the negative interest rate response to income changes under autarky in the case of $\sigma = 0$.

Consider a world economy where countries are inherently identical except for the initial income. Under autarky, there is a unique, symmetric, stable steady state where

¹³By explicitly taking into account the mass-of-entrepreneurs constraint, we correct an error in figure 5 of Matsuyama (2004). See the proof of Proposition 3 for the derivation of equations (28)-(30) and the threshold conditions for the borrowing constraints and the mass-of-entrepreneurs constraint.

¹⁴The detailed analysis and the proof of proposition 2 are included in the proof of proposition 3.

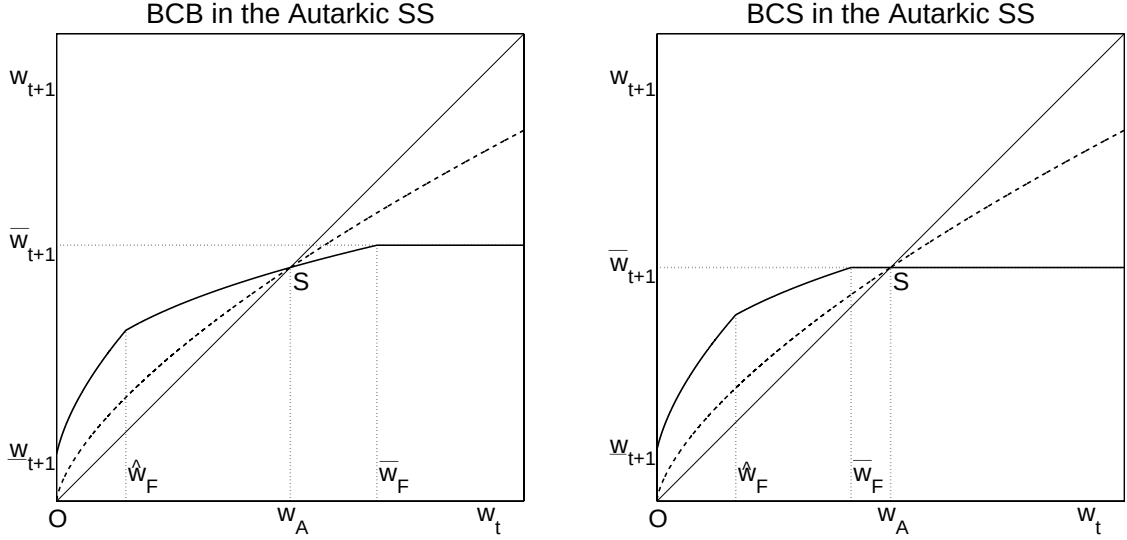


Figure 5: Laws of Motion for Wage: the Case $\sigma = 0$

all countries reach the same income level in the long run, regardless of their initial income levels. Financial globalization does not change the uniqueness and stability of this symmetric steady state in the world economy.

4.2 Dynamic and Steady-State Analysis: the Case of $\sigma = 1$

According to lemma 2, in the case of the constant MIR $\sigma = 1$, the autarkic interest rate increases in aggregate income, if the extensive-margin effect dominates the neoclassical effect. In contrast to the case of $\sigma = 0$, financial integration may generate the investment-amplification effect, which act as a divergence force and compete with the decreasing MPK, leading to multiple steady states in country i . The factors that can reinforce the positive interest rate response to income changes under autarky may raise the possibility of multiple steady states under financial integration.

Proposition 3. *In the case of $\sigma = 1$, let $\hat{\lambda}_F \equiv \frac{\alpha - \theta}{1 - \theta}$. Under financial integration, if $\theta < \alpha$ and $\lambda \in (0, \hat{\lambda}_F)$, multiple steady states may arise for an individual country; otherwise, the autarkic steady state is still the unique, stable steady state.*

Wealth Inequality and the Possibility of Symmetry Breaking

Figure 6 shows proposition 3 in the $\{\lambda, \theta\}$ space and the solid curve shows the threshold value $\hat{\lambda}_F$ as the function of θ . For parameters in region U, the autarkic steady state is still the unique, stable steady state under financial integration; for parameters in region M, financial integration may lead to multiple steady states for an individual country so that financial globalization may “break” the symmetric steady state under autarky and lead to income polarization in the world economy, as shown later.

Before exploring the mechanism behind the multiple steady states, we compare our results with Matsuyama (2004). In his model, agents have the homogeneous labor endowment so that their labor income is homogeneous and so is their wealth. His model is

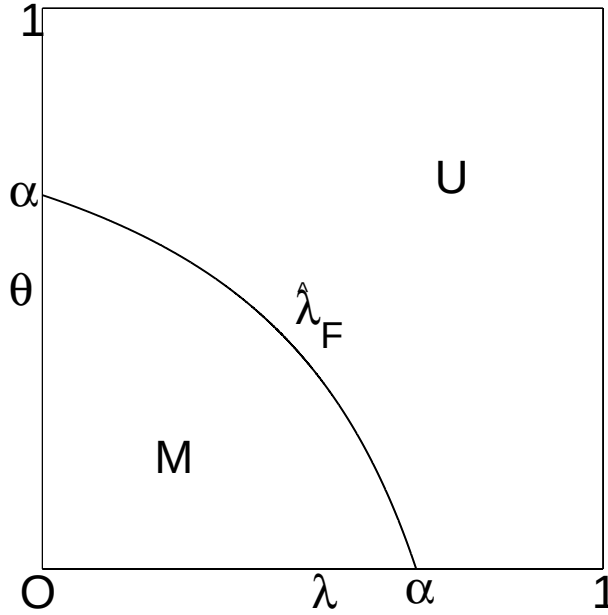


Figure 6: Possibility of Multiple Steady States in the $\{\lambda, \theta\}$ Space: $\sigma = 1$

technically a limiting case of ours with $\theta \rightarrow 0$.¹⁵ According to proposition 3 and figure 6, a necessary condition for symmetry breaking in his model is $\lambda < \alpha$,¹⁶ implying that the level of financial development is a key determinant for symmetry breaking.

In our model, the larger the θ , the greater the wealth inequality, the weaker the extensive-margin effect on the credit market, the less likely the autarkic interest rate is higher in the rich than in the poor country,¹⁷ the less likely financial globalization leads to "uphill" financial capital flows from the poor to the rich country, the less likely the cross-country income polarization. According to proposition 3 and figure 6,

- for $\theta \in (0, \alpha)$, a rise in wealth inequality reduces the possibility of symmetry breaking through the threshold value $\hat{\lambda}_F$;
- for $\theta \in (\alpha, 1)$, symmetry breaking does not arise at all, regardless of the level of financial development.

Thus, our result suggests that wealth inequality is as crucial as financial development in affecting the consequences of financial globalization.¹⁸

¹⁵See the proofs of propositions 3 and 6 in appendix.

¹⁶Suppose that the production function in his model is Cobb-Douglas with α as the capital share.

¹⁷See the discussion at the end of section 3.

¹⁸In our model, α represents the capital share in the Cobb-Douglas production function and the wealth distribution is Pareto with the tail index $\frac{1}{\theta}$ and the Gini coefficient $\frac{1}{\frac{1}{\theta}-1}$. Our model is highly stylized and not suitable for calibration. However, one can still infer θ using the Gini coefficients from the World Bank database and compare it with the conventional value for α to get a sense about whether multiple steady states are possible at all.

Exploring the Mechanism of Symmetry Breaking

Let us focus on the case of $\theta < \alpha$ where multiple steady states may arise. Define a composite parameter $Z \equiv \frac{R}{\rho} \left(\frac{1-\theta}{m} \right)^{\frac{1}{\rho}}$. The left (right) panel of figure 7 shows the parameter configuration for multiple steady states under financial integration in the $\{\lambda, \psi_A\}$ ($\{\lambda, Z\}$) space.¹⁹ The dashed line in the left panel shows the threshold value $\tilde{\psi}_A$ defined in lemma 2. The solid (dash) curves in figure 8 show the laws of motion for wage under financial integration (autarky), with the parameters in the five regions of figure 7, respectively.

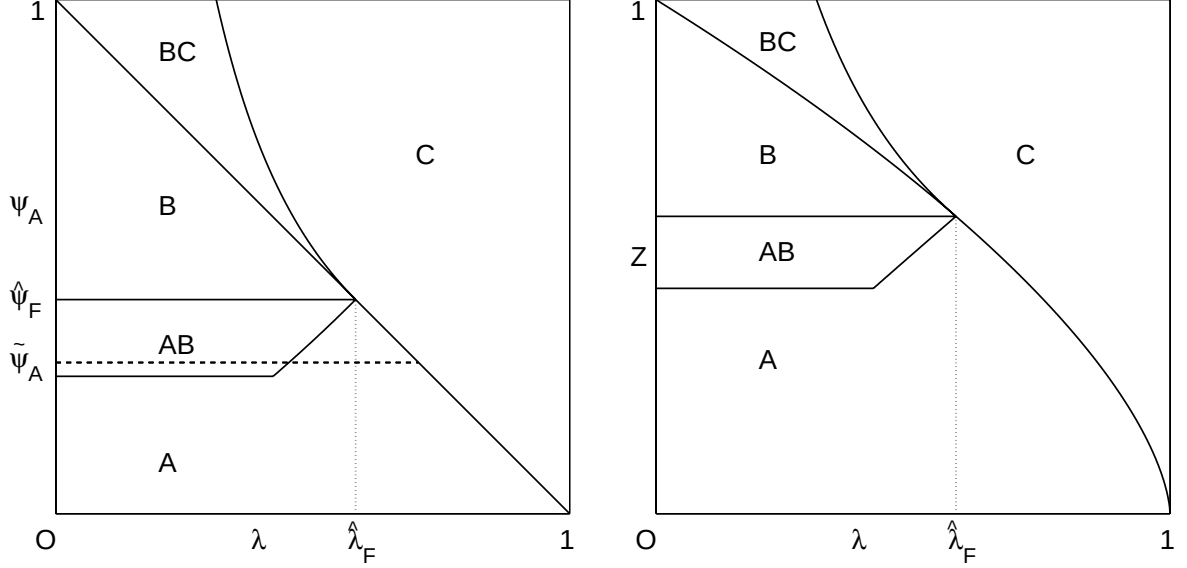


Figure 7: Parameter Configurations for Multiple Steady States: $\sigma = 1$ and $\theta < \alpha$

Consider the parameters in region B of figure 7. According to the upper-left panel of figure 8, financial integration destabilizes the autarkic steady state S in the sense that, with the initial income higher (lower) than in the autarkic steady state $w_0^i > w_A$ ($w_0^i < w_A$), country i converges to the steady state H (L) with $w_H > w_A$ ($w_L < w_A$).

Intuitively, for the parameters in region BI of figure 3, if aggregate income in country i is initially higher than in the autarkic steady state $Y_0^i > Y_A$, the autarkic interest rate is higher than the world level $r_0^i > r^*$, given $r^* = r_A$. In period $t = 0$, country i witnesses financial capital inflows, which makes the current domestic investment and next-period aggregate income higher than under autarky. The opposite applies to the case of $Y_0^i < Y_A$. Thus, due to the investment-amplification effect, the law of motion for wage around the autarkic steady state is steeper under financial integration than under autarky.

¹⁹According to equation (26), $\psi_A = Z^{\rho(1-\theta)}$ is a function of the model parameters and one can convert the boundary conditions from the $\{\lambda, \psi_A\}$ space (the left panel) to the $\{\lambda, Z\}$ space (the right panel). Matsuyama (2004) normalizes the FIR at unity and shows with figure 5 the parameter configuration in the $\{\lambda, R\}$ space. Then, he analyzes the impact of the productivity, R , on symmetry breaking. In our model, we introduce m as a free parameter for the MIR. Technically, both R and m matter for symmetry breaking. If the various combinations of R and m give the same value of Z , ψ_A is unaffected and so are the boundary conditions in the $\{\lambda, \psi_A\}$ and $\{\lambda, Z\}$ spaces. One can map the diagram from the $\{\lambda, Z\}$ space to the $\{\lambda, R\}$ space or to the $\{\lambda, m\}$ space to analyze the impacts of R or m , respectively. Matsuyama's model is a limiting case of ours. With $\sigma = 1$ and $\theta \rightarrow 0$, the right panel of figure 15 coincides with figure 5 in Matsuyama (2004). See appendix A and the proofs of propositions 3 and 6.

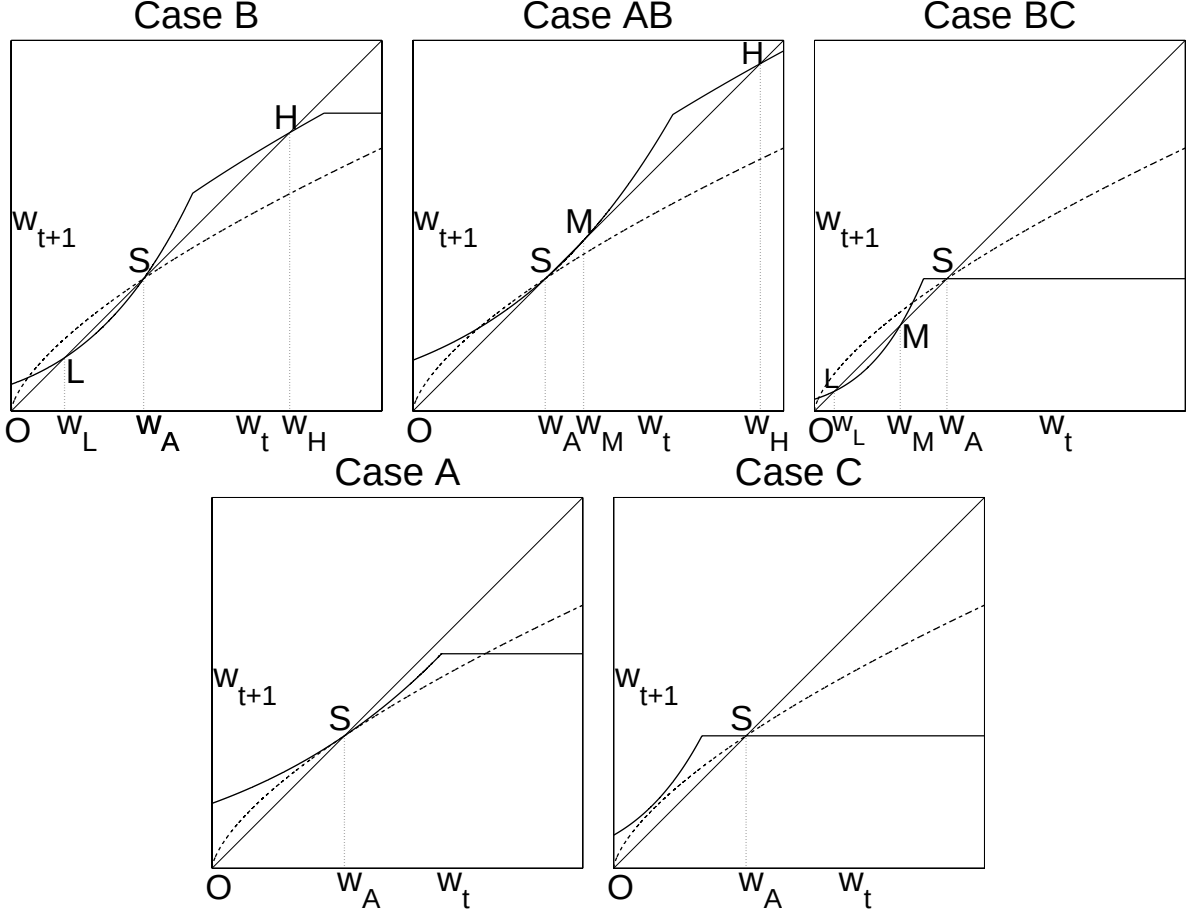


Figure 8: Laws of Motion for Wage: $\sigma = 1$ and $\theta < \alpha$

Given Y_0^i and Y_A , the larger the interest rate response to income changes around the autarkic steady state, the larger the interest rate deviation from the world level, the larger the magnitude of financial capital flows, the stronger the investment-amplification effect. As shown at the end of section 3, given the borrowing constraints are binding at the autarkic steady state, the larger the ψ_t , the larger the extensive-margin effect, the larger the positive interest rate response to income changes at the autarkic steady state. Thus, for the parameters in region B of figure 7, the interest rate response to income changes at the autarkic steady state is positive and sufficiently strong so that the investment-amplification effect dominates the decreasing MPK and hence, the autarkic steady state becomes unstable under financial integration.

Financial capital inflows in period $t = 0$ lead to higher aggregate income in period $t = 1$, which allows more agents to meet the MIR and become entrepreneurs. As long as the borrowing constraints are binding and the mass-of-entrepreneurs constraint is slack $\tau_t^i < 1$, the extensive-margin effect dominates the neoclassical effect so that the interest rate tends to rise in country i , which further attracts financial capital inflows in period $t = 1$. This way, financial capital flows become dynamically self-reinforcing, which magnifies the investment-amplification effect over time. This dynamic self-reinforcing process goes on until either the borrowing constraints become slack or the mass-of-entrepreneurs constraint becomes binding. Here, the extensive-margin channel is critical to the investment-amplification effect in the short run and in the long run.

Start with region B of figure 7. Let us raise $\mathbf{m}$ so that ψ_A declines and the parameter configuration moves into region AB. The interest rate response to income changes at the autarkic steady state is positive but small. Thus, the decreasing MPK dominates the investment-amplification effect so that the autarkic steady state is still locally stable under financial integration. However, for $w_t^i \gg w_A$, ψ_t^i enters into the upper part of region BI of figure 3 where the interest rate response to income changes is positive and strong. Then, financial integration affects domestic investment and aggregate income in the same way as for case B. The upper-middle panel of figure 8 shows that, besides the stable steady state S, financial integration leads to another stable steady state H and an unstable steady state M with $w_H > w_M > w_A$.

Start with region B of figure 7. Let us reduce $\mathbf{m}$ so that ψ_A rises and the parameter configuration moves into region BC where the borrowing constraints are slack and the interest rate declines in aggregate income at the steady state under autarky. Given $r^* = r_A = \rho$, financial integration makes the law of motion for wage flat at the autarkic steady state and hence, the autarkic steady state is still locally stable. However, for $w_t^i \ll w_A$, ψ_t^i enters into region BI of figure 3 where the interest rate responds positively to income changes. Then, financial integration affects domestic investment and aggregate income in the same way as for case B. The upper-right panel of figure 8 shows that, besides the stable steady state S, financial integration leads to another stable steady state L and an unstable steady state M with $w_L < w_M < w_A$.

For the parameters in region A and C of figure 7, financial integration does not lead to multiple steady states. See the lower panels of figure 8.

To sum up, in the case of $\sigma = 1$, the positive interest rate response to income changes under autarky is key to the rise of multiple steady states under financial integration. As shown in section 3, five factors affect the response of the autarkic interest rate to income changes, which explains intuitively the threshold conditions specified in proposition 3.

Although the initial income level does not matter for the dynamic patterns and the steady state in country i under autarky, it may matter under financial integration. For example, in case B, starting with the income level *slightly* higher (lower) than that in the autarkic steady state, the small open economy converges to a new, stable steady state with the income level much higher (lower) than in the autarkic steady state; in case AB (BC), starting with the income level *sufficiently* higher (lower) than that in the autarkic steady state, the small open economy converges to a new, stable steady state with the income level much higher (lower) than in the autarkic steady state.

Incorporate this mechanism into a world economy where all countries are inherently identical. For the parameters in region B of figure 7, the world economy has a unique, stable, symmetric steady state under autarky where all countries reach the same income level in the long run, regardless of their initial income levels; financial globalization may destabilize this symmetric steady state and lead to the asymmetric, stable steady state where initially rich (poor) countries have the income higher (lower) than in the autarkic steady state. Matsuyama (2004) calls this result “symmetry breaking”.

5 Heterogeneity in Financial Development

So far, we have assumed that countries are inherently identical except for the initial income so as to highlight the mechanism behind “symmetry breaking”. Needless to say, countries differ in various aspects, e.g., endowments, technologies, institutions, etc. In this section, we introduce the cross-country heterogeneity in financial development²⁰ and address three issues for a small open economy, i.e., the threshold conditions for financial integration, domestic financial crises and capital controls, and the world interest rate shocks and economic volatility.

We conduct our analysis in a two-country setting of the constant MIR ($\sigma = 1$). The world consists of country S (small) and country W (the rest of world) which are inherently identical except for the level of financial development, $\lambda^S \neq \lambda^W$, and the population size, $L^S \ll L^W$. Suppose that country W allows free mobility of financial capital and is initially in the steady state. Given its negligible population size in the world economy, country S takes the world interest rate as given, $r_t^* = r_A^W$. Note that ψ_A is a function of the parameters $\{R, \rho, \theta, \mathbf{m}\}$, identical in the two countries.

Define $\Lambda^W \equiv \frac{(1-\alpha)(1-\theta)}{(1-\lambda^W)^{\frac{1}{1-\theta}}}$ as a function of $\lambda^W \in (0, 1)$ and $\frac{\partial \Lambda^W}{\partial \lambda^W} > 0$.

Corollary 1. *Under autarky, if $\mathbf{m} > \left(\frac{R}{\rho}\right)^\rho \Lambda^W$, the borrowing constraints in country W are binding in the steady state with $\psi_A \in (0, 1 - \lambda^W)$ and $r_A^W = \frac{\lambda^W}{1-\psi_A} \rho < \rho$.*

For $\lambda^S \in (0, 1 - \psi_A)$, the borrowing constraints in country S are binding in the autarkic steady state with $r_A^S = \frac{\lambda^S}{1-\psi_A} \rho < \rho$; for $\lambda^S \in (1 - \psi_A, 1)$, the borrowing constraints in country S are slack in the autarkic steady state with $r_A^S = \rho$.

We focus on the case where the parameters for country W is in region B of figure 7.

5.1 Threshold Conditions for Financial Integration

A country will choose to be financially integrated if and only if it is beneficial.²¹ The condition of financial integration can be quite different for those financially developed versus those financially underdeveloped countries.

As shown in subsection 4.2, for financially developed countries (i.e., $\lambda > \hat{\lambda}_F$), financial integration does not lead to symmetry breaking and it may speed up the convergence for an individual country to the autarkic steady state. Thus, these countries are more likely to have the integrated financial markets.

In the following, we focus on the financially underdeveloped countries.

Proposition 4. *Given (λ^W, ψ_A) in region B of figure 7,²² there are two threshold values, $\underline{\lambda}^S < \lambda^W$ and $\bar{\lambda}^S > \lambda^W$.*

²⁰Alternatively, one can also introduce the cross-country heterogeneity in wealth inequality and analyze the implications of financial integration.

²¹For simplicity, the per capita income is used here as a measure of welfare. For a formal welfare analysis, one needs to take into account the distributional effects on entrepreneurs and on households, respectively.

²²By the same logic, one can also analyze the cases with (λ^W, ψ_A) in the other four regions.

- For $\lambda^S < \underline{\lambda}^S$, financial integration induces country S to converge to a unique steady state with aggregate income lower than under autarky;
- for $\lambda^S \in (\underline{\lambda}^S, \bar{\lambda}^S)$, financial integration leads to multiple steady states for country S . Let $\tilde{w}^S$ denote the wage rate in the unstable steady state. Financial integration induces country S to converge to the steady state with aggregate income higher (lower) than under autarky, iff its initial income is sufficiently high (low), i.e., $w_0^S > \tilde{w}^S$ ($w_0^S < \tilde{w}^S$);
- for $\lambda^S > \bar{\lambda}^S$, financial integration induces country S to converge to a unique steady state with aggregate income higher than under autarky.

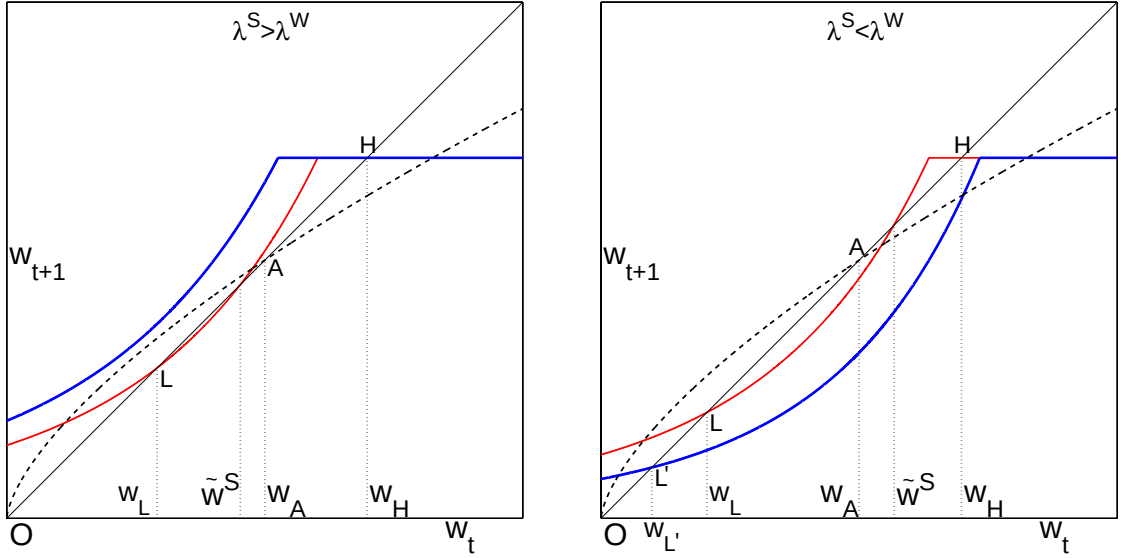


Figure 9: Impacts of Financial Integration: the Case of $\sigma = 1$

Let us start with the case of $\lambda^S > \lambda^W$. The dashed curve in the left panel of figure 9 shows the law of motion for wage in country S under autarky. If country S is initially at the autarkic steady state $Y_0^S = Y_A$, the interest rate is higher in country S than in country W , $r_A^S = \min\{\frac{\lambda^S}{1-\psi_A}\rho, \rho\} > r_A^W = \frac{\lambda^W}{1-\psi_A}\rho$, due to the financial-development effect. From period $t = 0$ on, country S receives financial capital inflows, which raises its current domestic investment and next-period aggregate income. According to the mechanism described in subsection 4.2, the investment-amplification effect dominates the decreasing MPK so that aggregate income in country S rises over time until country S reaches the new steady state. According to proposition 4,

- if λ^S is significantly higher than λ^W , i.e., $\lambda^S > \bar{\lambda}^S$, the law of motion for wage under financial integration is shown by the thick curve in the left panel of figure 9. Regardless of the initial income level, country S converges to the unique steady state H with the income much higher than under autarky.
- If λ^S is slightly higher than λ^W , i.e., $\lambda^S \in (\lambda^W, \bar{\lambda}^S)$, the law of motion for wage under financial integration is shown by the thin curve in the left panel of figure 9. If its initial income level is relatively low (high) $w_0^S < \tilde{w}^S$ ($w_0^S > \tilde{w}^S$), country

S converges to the steady state L (H) with the income much lower (higher) than under autarky. The higher the λ^S , the lower the threshold value $\tilde{w}^S$.

The analysis for the case of $\lambda^S < \lambda^W$ is analogue to the previous one. According to proposition 4,

- if λ^S is significantly lower than λ^W , i.e., $\lambda^S < \underline{\lambda}^S$, the law of motion for wage under financial integration is shown by the thick curve in the right panel of figure 9. Regardless of the initial income level, country S converges to the unique steady state L' with the income much lower than under autarky.
- If λ^S is slightly lower than λ^W , i.e., $\lambda^S \in (\underline{\lambda}^S, \lambda^W)$, the law of motion for wage under financial integration is shown by the thin curve in the right panel of figure 9. If its initial income level is relatively low (high) $w_0^S < \tilde{w}^S$ ($w_0^S > \tilde{w}^S$), country S converges to the steady state L (H) with the income much lower (higher) than under autarky. The higher the λ^S , the lower the threshold value $\tilde{w}^S$.

Proposition 4 essentially provides the threshold conditions for financial integration: a country should (should not) allow financial integration if its level of financial development is far above (below) the world level; in the intermediate case, the initial income matters and, the higher the level of financial development, the lower the threshold value for the initial income, i.e., $\tilde{w}^S$. Thus, whether a country should allow financial integration depends on its level of financial development relative to the rest of the world (country W) and possibly on its initial income level.

In order to highlight the extensive-margin effect due to the constant MIR and financial frictions, we conduct a parallel analysis in the setting where the project is not subject to the MIR. See appendix A.4.1. In that setting, aggregate investment only takes place on the intensive margin and the autarkic interest rate strictly declines in aggregate income. Thus, financial integration does not lead to multiple steady states and the steady-state income effect is much smaller than in the setting of the constant MIR $\sigma = 1$.

Kose, Prasad, Rogoff, and Wei (2009, 2010) and Kose, Prasad, and Taylor (2011) provide the empirical evidences that financial integration leads to better macroeconomic outcomes when certain threshold conditions are met. The extent of financial sector development is one of their threshold conditions and hence, our model predictions are consistent with their findings.

As shown above, the condition of financial integration is quite different for those financially developed versus those financially underdeveloped countries. It is consistent with the fact that the developed countries which have the well-developed financial sector, e.g., the OECD countries, usually promote capital account liberalization at the global level, while emerging economies which have the underdeveloped financial sector, e.g., China and India, are rather conservative on this issue.

5.2 Domestic Financial Crises and Capital Controls

In this subsection, we feature a domestic financial crisis in country S by a decline in its level of financial development from λ_1^S to λ_2^S in period $t = 0$ and analyze its short-run and the long-run impacts as well as discuss the conditions for capital controls.

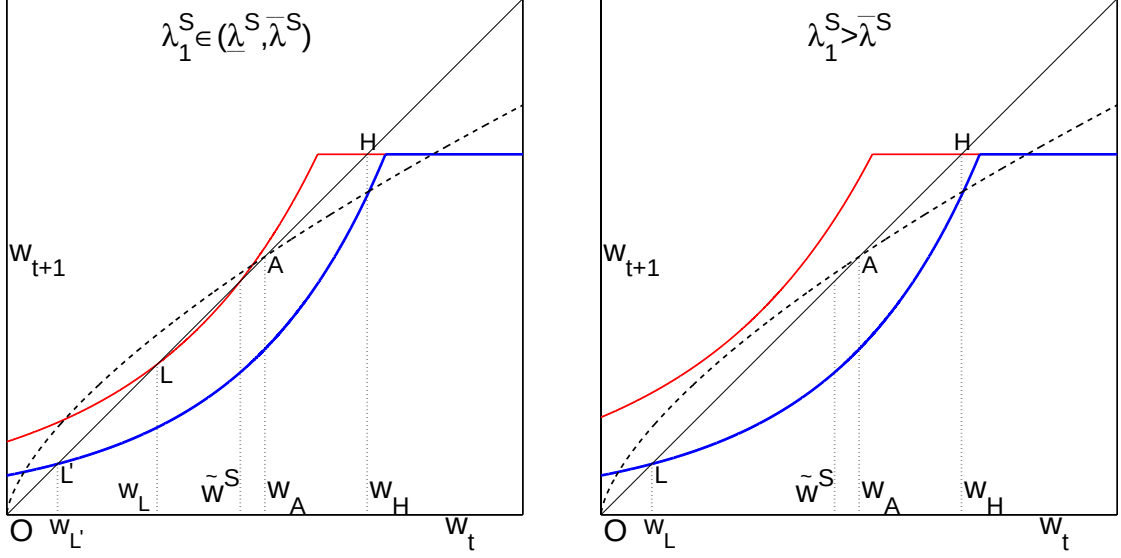


Figure 10: Impacts of Domestic Financial Crisis: the Case of $\sigma = 1$

Under autarky, as domestic saving is fully used to produce capital goods, the decline in the level of financial development in country S does not affect the law of motion for wage, as shown by the dashed curve in figure 10.

Consider the equilibrium under financial integration in the scenario of $\lambda_1^S \in (\underline{\lambda}^S, \bar{\lambda}^S)$. The thin curve in the left panel of figure 10 shows the law of motion for wage before period $t = 0$. Suppose that country S is initially in the steady state H with $w_0^S = w_H$.²³ A decline in λ^S shifts the law of motion for wage to the right and the impacts on aggregate dynamics and on the steady state depends on its magnitude and duration.

- For a small decline, i.e., $\lambda_2^S \in (\underline{\lambda}^S, \lambda_1^S)$, the law of motion for wage shifts slightly to the right and point H is still a stable steady state. Since the equilibrium allocation is unaffected in this scenario, it is not considered as a financial crisis.
- For a large decline, i.e., $\lambda_2^S < \underline{\lambda}^S$, the law of motion for wage shifts significantly to the right so that point H is no longer a steady state, as shown by the thick curve. Intuitively, the tightened borrowing constraints depress the domestic credit demand and the interest rate declines, which triggers financial capital outflows and the fall in domestic investment. This scenario can be considered as a financial crisis.
 - If λ^S falls temporarily and then quickly returns to its initial level before w_t^S falls below $\tilde{w}^S$, country S eventually returns to its initial steady state H.
 - For a prolonged decline in λ^S , country S eventually converges to the steady state L'. In that case, even if λ^S returns to the pre-crisis level, country S is trapped in the steady state L with the income lower than under autarky.

²³According Proposition 4, for $\lambda_1^S \in (\underline{\lambda}^S, \bar{\lambda}^S)$, if the initial income in country S is below a threshold, financial integration induces country S to converge to the steady state L with the income lower than under autarky. In that case, country S should choose autarky.

Consider the equilibrium under financial integration in the scenario of $\lambda_1^S > \bar{\lambda}^S$.²⁴ The thin curve in the right panel of figure 10 shows the law of motion for wage before period $t = 0$. Suppose that country S is initially in the steady state H, $w_0^S = w_H$. Similar as in the previous case, the impacts of a decline in λ^S on aggregate dynamics and on the steady state depends on its magnitude and duration; different from the previous case, even for a large and prolonged decline in λ^S , country S can always return to the initial steady state H when λ^S returns to its pre-crisis level.

To sum up, for a mild and temporary disruption in the domestic financial market, country S does not need to impose capital controls, because the economy can recover to its initial level when the financial sector resumes its functions. However, in the case of a severe and prolonged domestic financial crisis, the less financially developed countries should impose capital controls to avoid the economy to be trapped in a low-income steady state, while the more financially developed countries do not need to do so. Intuitively, the existence of multiple steady states creates path dependence so that the timing and the size of policy interventions are critical for the less financially developed country to recover from domestic financial crises. Recently, the International Monetary Fund (hereafter, IMF) endorsed a new “institutional view” on capital account liberalization and the management of capital flows (IMF, 2012a,b, 2015). Capital outflow management measures can be introduced temporarily in crisis situations or when a crisis may be imminent. Our results are consistent with the IMF’s position.

5.3 World Interest Rate Shocks and Economic Volatility

So far, we have assumed that country W is in the steady state and the world interest rate is constant at $r_t^* = r_A^W$. Suppose that country W is hit by a financial crisis in period $t = 0$ on and its level of financial development falls from λ_1^W to λ_2^W . Accordingly, the interest rate in country W falls gradually from $r_1^W = \frac{\lambda_1^W \rho}{1-\psi_W}$ to $r_2^W = \frac{\lambda_2^W \rho}{1-\psi_W}$ and so does the world interest rate. As shown below, the impacts of the world interest rate shock on country S depend on λ^S and the size of the world interest rate change. Note that $\bar{\lambda}^S$ and $\underline{\lambda}^S$ refer to the two threshold values defined in proposition 4, where $\bar{\lambda}^S > \lambda_1^W > \underline{\lambda}^S$.²⁵

- For $\lambda^S > \bar{\lambda}^S$, the thin (thick) curve in the left panel of figure 11 shows the law of motion for wage in country S under financial integration before (after) period $t = 0$. Before period $t = 0$, country S optimally chooses to allow financial integration, according to proposition 4. Suppose that country S is initially in the steady state with the wage rate $w_H = w_A \left(\frac{\rho}{r_1^W} \right)^\rho = w_A \left(\frac{1-\psi_W}{\lambda_1^W} \right)^\rho > w_A$. From period $t = 0$ on, the decline in the world interest rate allows country S to further receive financial capital inflows and it eventually converges to a new steady state H’ with $w_{H'} = w_A \left(\frac{\rho}{r_2^W} \right)^\rho = w_A \left(\frac{1-\psi_W}{\lambda_2^W} \right)^\rho$.

²⁴According to Proposition 4, for $\lambda_1^S < \bar{\lambda}^S$, financial integration reduces the steady-state aggregate income in country S and hence, country S optimally chooses autarky.

²⁵As shown in the proof of proposition 4, the two threshold values are the functions of λ^W . In this subsection, we use λ_1^W to compute the two threshold values.

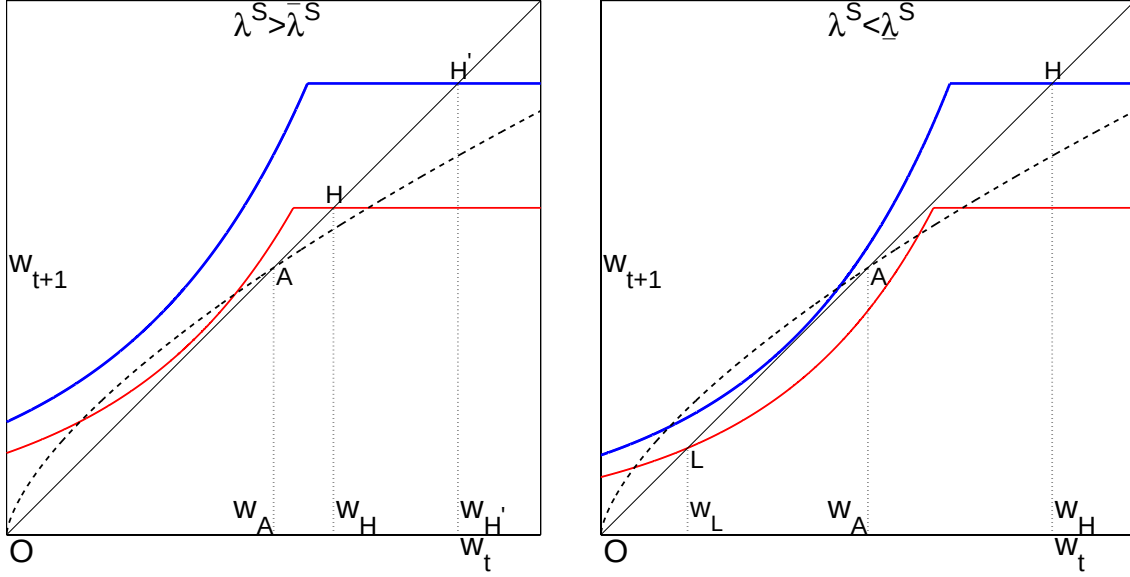


Figure 11: Impacts of the World Interest Rate Change: the Case of $\sigma = 1$

- For $\lambda^S < \bar{\lambda}^S$, the thin (thick) curve in the right panel of figure 11 shows the law of motion for wage in country S under financial integration before (after) period $t = 0$. Before period $t = 0$, country S optimally chooses autarky, according to proposition 4. Suppose that country S is initially in the steady state with the wage rate w_A . From period $t = 0$ on, if the world interest rate falls below the domestic interest rate in country S $r_2^W < r_A^S$ or equivalently $\lambda_2^W < \lambda^S$, country S optimally chooses financial integration and it eventually converges to a new steady state H with $w_H = w_A \left(\frac{\rho}{r_2^W} \right)^\rho = w_A \left(\frac{1-\psi_W}{\lambda_2^W} \right)^\rho$.

Suppose that the level of financial development in country W gradually returns to its initial level λ_1^W and the world interest rate also rises to its initial level r_1^W . By the same logic as mentioned above, the wage rate in the more financially developed country (i.e., $\lambda^S > \bar{\lambda}^S$) declines from $w_{H'}$ to w_H and so does the aggregate income; the wage rate in the less financially developed country (i.e., $\lambda^S < \bar{\lambda}^S$) declines from w_H to w_L if it does not restrict financial capital flows or from w_H to w_A if it imposes capital controls again. To sum up, the world interest rate shocks generate a larger income volatility in the less financially developed country than in the more financially developed country. This is consistent with our observations of financial capital flows for emerging economies in the period of quantitative easing as well as in the post-QE period.

6 Final Remarks

In the literature, finance plays a central role in theories of wealth inequality (Aghion and Bolton, 1997; Banerjee and Newman, 1993; Demircuc-Kunt and Levine, 2009; Galor and Moav, 2004; Galor and Zeira, 1993; Greenwood and Jovanovic, 1990). The empirical evidence is fairly robust that financial development improves the distribution of income and especially the incomes of the poor (Beck, Demircuc-Kunt, and Levine, 2007; Burgess

and Pande, 2005). Meanwhile, recent evidence suggests that poor access to finance not only reflects economic constraints but also barriers erected by insiders. Inequality affects the distribution of political influence, so financial regulation often is easily captured by established interests in unequal countries (Claessens and Perotti, 2007; Demirguc-Kunt and Levine, 2009). In our paper, both the degree of wealth inequality and the level of financial development are exogenous²⁶, which allows us to isolate their respective impacts on symmetry breaking and obtain the analytical results. Our current findings suggest that the higher the wealth inequality and/or the higher the level of financial development, the less likely the symmetry breaking. Allowing financial development and wealth distribution to be jointly determined may have profound implications for symmetry breaking.

In this paper, we analyze how wealth inequality may reshape the consequences of financial globalization. Jaumotte, Lall, and Papageorgiou (2013) show that financial globalization is associated with a rise in inequality. Another possible extension of the current paper is to study how the dynamic interactions between wealth distribution and financial globalization may reshape the mechanism of symmetry breaking.²⁷

As is well known, the financial sectors not only promote economic growth and development (Buera, Kaboski, and Shin, 2011; King and Levine, 1993; Levine, Loayza, and Beck, 2000) but also evolve endogenously along the process of economic development (Acemoglu and Zilibotti, 1997; Laeven, Levine, and Michalopoulos, 2015; Song and Thakor, 2010). In general, the financial sectors are more efficient in the rich than in the poor countries. In appendix B.1, we introduce into our model the endogeneity of financial and economic development in a parsimonious way and its impacts on symmetry breaking are ambiguous. For future research, we plan to investigate how economic development may reshape the financial sector and then affect the symmetry breaking mechanism.

Our results suggest that countries should improve their financial sectors so as to reap the benefits from financial integration. However, should a country improve its financial sector before liberalizing its capital account or should it use capital account liberalization as a commitment device to stimulate the domestic financial intermediaries to improve their services? In order to answer this question, one must explicitly characterize the incentive structures of the financial intermediaries and investigate how they respond to financial integration. According to Alessandria and Qian (2005) and Tressel and Verdier (2011), financial integration may improve or worsen the efficiency of financial intermediaries, leading to an improvement or worsening of the aggregate composition of investment projects. In particular, their results depend critically on the institutional environment. Thus, it requires a more fundamental approach from the political economy perspective if one wants to analyze whether and how financial openness may trigger the changes in the institutional environment. We keep them for future research.

²⁶In our model, the labor income is the only source of the net wealth of young agents. Thus, their wealth distribution is identical as their labor income distribution, which is purely determined by the exogenous labor endowment distribution.

²⁷One can extend our model into a three-period setting where agents can accumulate wealth over time and then analyze the interactions between wealth inequality and financial globalization.

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Appendix

A FIR and Symmetry Breaking: Matsuyama (2004)

In this section, we replicate the results of Matsuyama (2004) and highlight the role of the extensive-margin channel by comparing the results in the settings with and without the FIR. The setting with the FIR is essentially a limiting case of our model with the constant MIR $\sigma = 1$ and $\theta \rightarrow 0$.

A.1 Model Settings with and without the FIR

The two model settings are identical to that in sector 2 except for two aspects:

- all agents have the same labor endowment which is normalized at one, and
- the investment project is not subject to the MIR.

In the first setting, agents differ in the technology endowment. A fraction of agents born in period t and in country i have the linear project to convert m_t^i units final goods in period t to $k_{t+1}^i = Rm_t^i$ units of capital goods in period $t + 1$, and they are called entrepreneurs; with no project, other agents only lend out their labor incomes and are called households. The mass of entrepreneurs $\tau \in (0, 1)$ is exogenous, while the entrepreneurial project size m_t^i is endogenous. Aggregate investment takes place on the *intensive margin*, $K_{t+1}^i = \tau Rm_t^i$. With **no** investment size requirement, it is called setting **N**.

In the second setting, all agents born in period t and country i are equally endowed with the indivisible project to convert $\mathbf{m}$ units of final goods in period t into $R\mathbf{m}$ units of capital goods in period $t + 1$.²⁸ If $w_t^i < \mathbf{m}$, an agent must borrow $\mathbf{m} - w_t^i$ to start its project, but the aggregate saving is not sufficient to allow all agents to run their projects. According to Matsuyama (2004), random credit rationing allows a fraction $\tau_t^i \in (0, 1)$ of agents to get the loan $\mathbf{m} - w_t^i$ to run the projects and they are called entrepreneurs, while others can only lend out their labor incomes and are called households. Different from setting **N**, the entrepreneurial project size $\mathbf{m}$ is exogenous, while the mass of entrepreneurs τ_t^i is endogenous.²⁹ Thus, aggregate investment takes place only on the *extensive margin*, $K_{t+1}^i = \tau_t^i R\mathbf{m}$. With **fixed** investment size requirement, it is called setting **F**.

Figure 12 shows the individual production function in the two settings. In setting **N**, the individual project is linear, $k_{t+1}^i = Rm_t^i$. In setting **F**, the project output is constant at $R\mathbf{m}$ for the input $m_t^i \geq \mathbf{m}$ and is zero for $m_t^i \in [0, \mathbf{m})$. For simplicity, we use τ_t^i and m_t^i to denote the mass of entrepreneurs and the individual investment size in the model description. Setting **N** is characterized by the fixed mass of entrepreneurs, $\tau_t^i = \tau$, while setting **F** is characterized by the fixed project size, $m_t^i = \mathbf{m}$.

Entrepreneurs are subject to the borrowing constraints (3). Since households are homogeneous and so are entrepreneurs, we drop the subscript j .

²⁸Matsuyama (2004) implicitly normalizes the individual project size at $\mathbf{m} = 1$, while we allow $\mathbf{m}$ to be a free parameter and analyze its impacts on symmetry breaking.

²⁹Although the FIR results in the non-convexity of the individual production set, Matsuyama (2007, 2008) argues that assuming a continuum of homogeneous agents convexifies the production set.

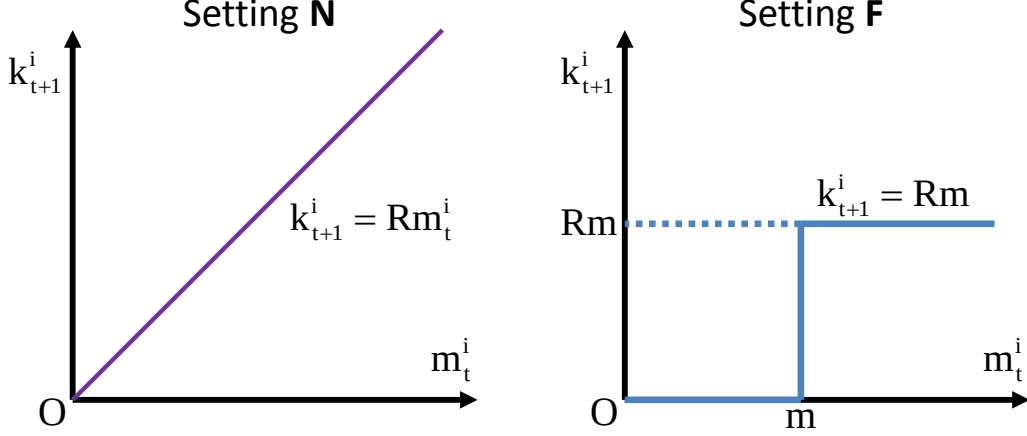


Figure 12: Individual Projects in the Two Settings

Under autarky, the markets for capital goods and credit clear each period.

$$K_{t+1}^i = \tau_t^i R m_t^i, \quad (\text{A.1})$$

$$\tau_t^i (m_t^i - w_t^i) = (1 - \tau_t^i) w_t^i. \quad (\text{A.2})$$

Definition 3. Under autarky, a market equilibrium in country i is a set of agents' choices, $\{m_t^i, c_t^{i,e}, c_t^{i,h}, \psi_t^i\}$, the mass of entrepreneurs $\{\tau_t^i\}$, the prices $\{q_t^i, w_t^i, r_t^i, \Omega_t^i\}$, and aggregate quantities $\{Y_t^i, K_t^i\}$, satisfying equations (1)-(2), (6)-(8), (10)-(11), (A.1)-(A.2).

$\tau_t^i = \tau$ is exogenous in setting **N**, while $m_t^i = \mathbf{m}$ is exogenous in setting **F**.

Under financial globalization, the equilibrium conditions are identical as under autarky except for the domestic and world credit market conditions.

$$\tau_t^i (m_t^i - w_t^i) = (1 - \tau_t^i) w_t^i - \phi_t^i w_t^i, \quad (\text{A.3})$$

$$\int_0^1 w_t^i \phi_t^i di = 0. \quad (\text{A.4})$$

Definition 4. Under financial globalization, a market equilibrium in country i is a set of agents' choices, $\{m_t^i, c_t^{i,e}, c_t^{i,h}, \psi_t^i\}$, the mass of entrepreneurs $\{\tau_t^i\}$, the prices $\{q_t^i, w_t^i, \Omega_t^i\}$, and aggregate quantities $\{Y_t^i, K_t^i\}$, satisfying equations (1)-(2), (6)-(8), (10)-(11), (A.1), and (A.3), and the interest rate is equalized across countries $r_t^i = r_t^*$ and the world interest rate r_t^* is determined by equation (A.4).

$\tau_t^i = \tau$ is exogenous in setting **N**, while $m_t^i = \mathbf{m}$ is exogenous in setting **F**.

A.2 Equilibrium under Autarky

For simplicity, we suppress the country index i for the scenario of autarky.

In setting **N**, according to equations (A.1) and (A.2), the equity-investment ratio is constant at $\psi_t = \frac{w_t}{m_t} = \tau$ and domestic investment is fully financed by domestic saving $K_{t+1} = R\tau m_t = R w_t$. The law of motion for wage is characterized by equation (16).

In setting **F**, for $w_t < \mathbf{m}$, aggregate saving is too low to allow all agents to run their projects. According to equations (A.1) and (A.2), the mass of entrepreneurs and the equity-investment ratio are endogenous, $\tau_t = \psi_t = \frac{w_t}{\mathbf{m}} < 1$, and domestic investment is fully financed by domestic saving $K_{t+1} = R\tau_t \mathbf{m} = R w_t$. Then, the law of motion for wage is the same as in setting **N**.

For $w_t \geq \mathbf{m}$, all agents self-finance their projects, $\tau_t = \psi_t = 1$. Given the fixed project size, the aggregate output of capital goods is constant at $K_{t+1} = R\mathbf{m}$ and the law of motion for wage is flat at $w_{t+1} = \left(\frac{R\mathbf{m}}{\rho}\right)^\alpha$.

Proposition 5. *In setting $\mathbf{N}$, $\psi_t = \tau_t$; in setting $\mathbf{F}$, $\psi_t = \tau_t = \frac{w_t}{\mathbf{m}}$.*

In both settings, there exists a unique, stable steady state in each country under autarky;

iff $\psi_t \in (0, 1 - \lambda]$, the borrowing constraints are binding;

iff $\psi_t \in (1 - \lambda, 1]$, the borrowing constraints are slack.

Proposition 5 is essentially the same as proposition 1 and lemma 1.

Interest Rate Response to Income Changes

If the borrowing constraints are slack, $r_t = Rq_{t+1}$ and, according to equation (20), the interest rate declines in aggregate income.

Let us focus on the case of the binding borrowing constraints.

In setting $\mathbf{N}$, combine equations (6), (16), (A.1)-(A.2) to get

$$r_t = \frac{\lambda}{1 - \psi_t} q_{t+1} R = \frac{\lambda}{1 - \tau_t} w_t^{-(1-\alpha)} R^\alpha \rho^{1-\alpha}. \quad (\text{A.5})$$

With the extensive margin mute, the interest rate declines in aggregate income, due to the neoclassical effect, as in our model with $\sigma = 0$.

In setting $\mathbf{F}$, combine equations (6), (16), (A.1)-(A.2) to get

$$r_t = \frac{\lambda}{1 - \psi_t} q_{t+1} R = \frac{\lambda}{1 - \tau_t} w_t^{-(1-\alpha)} R^\alpha \rho^{1-\alpha}. \quad (\text{A.6})$$

Higher aggregate income raises the agents' labor income and the decline in the required loan size $\mathbf{m} - w_t$ allows more agents to get the loans and invest as entrepreneurs $\tau_t = \frac{w_t}{\mathbf{m}}$. The decline in the mass of households $(1 - \tau_t)$ reduces the aggregate credit supply on the extensive margin. If the extensive-margin effect dominates the neoclassical effect, the interest rate rises in aggregate income, as in our model with $\sigma = 1$.

Lemma 3. *Under autarky, the interest rate declines in aggregate income in setting $\mathbf{N}$; for $\lambda \in (0, \tilde{\lambda}_A)$, the interest rate rises in aggregate income in setting $\mathbf{F}$, iff $w_t \in (\tilde{\psi}_A \mathbf{m}, \bar{\psi} \mathbf{m})$ or equivalently $\psi_t = \frac{w_t}{\mathbf{m}} \in (\tilde{\psi}_A, \bar{\psi})$, where $\tilde{\psi}_A \equiv \frac{1-\alpha}{2-\alpha}$ and $\tilde{\lambda}_A \equiv 1 - \tilde{\psi}_A$.*

Figure 13 shows lemma 3 in the $\{\lambda, \psi_t\}$ space for setting $\mathbf{F}$, which is essentially the limiting case of figure 3 with $\theta \rightarrow 0$. The analysis follows that in section 3.

A.3 Equilibrium under Financial Globalization

From period $t = 0$ on, agents in country i are allowed to borrow or lend abroad. As a small open economy, country i takes the world interest rate as given $r_t^i = r^*$. Without loss of generality, we assume that $r^* = r_A$, where $r_A = \frac{\lambda}{1-\psi_A} \rho$ ($r_A = \rho$) if the borrowing constraints are binding (slack) in the autarkic steady state.

Proposition 6. *Under financial integration, the autarkic steady state is still the unique, stable steady state in setting $\mathbf{N}$, while multiple steady states may arise in setting $\mathbf{F}$ if $\lambda \in (0, \hat{\lambda}_F)$, where $\hat{\lambda}_F \equiv \alpha$.*

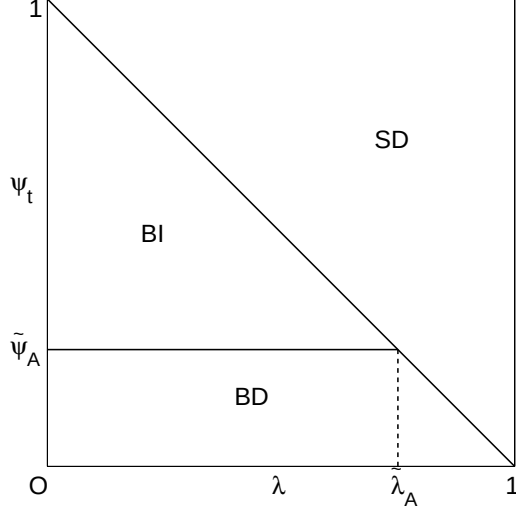


Figure 13: Interest Rate Patterns in the $\{\lambda, \psi_t\}$ Space under Autarky

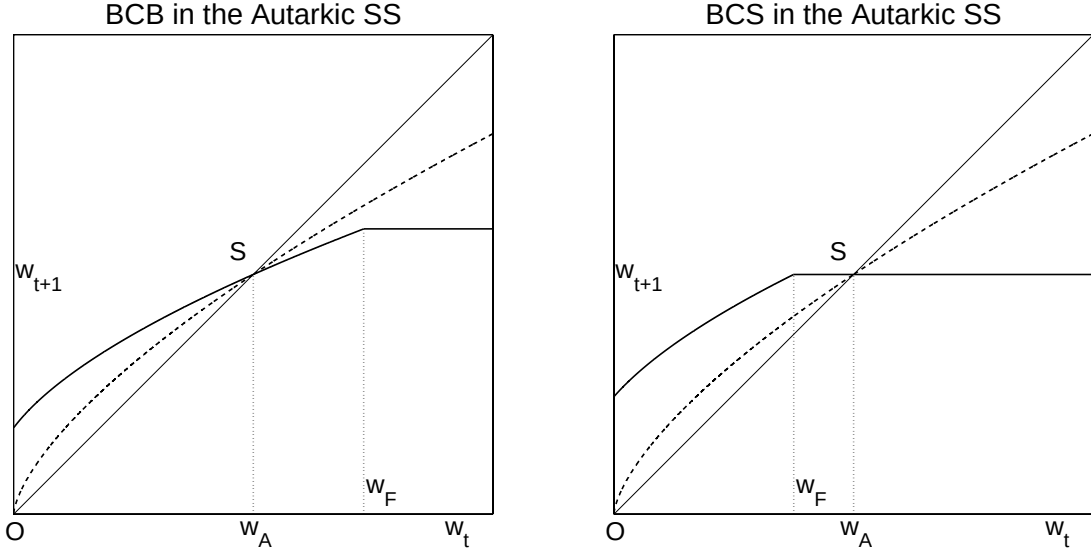


Figure 14: Laws of Motion for Wage: Setting N

Proposition 6 is essentially the limiting case of proposition 3 with $\theta \rightarrow 0$.

In setting N, the autarkic interest rate declines in aggregate income. As in our model with $\sigma = 0$, financial integration generates the investment-dampening effect, which makes the law of motion for wage around the autarkic steady state flatter than under autarky. The solid (dashed) curves in figure 14 show the laws of motion for wage under financial integration (autarky). The left (right) panel shows the case where the borrowing constraints are binding (slack) in the steady state.

In setting F, the autarkic interest rate may rise in aggregate income. As in our model with $\sigma = 1$, financial integration may generate the investment-amplification effect, which competes with the decreasing MPK. Let $Z \equiv \frac{R}{\rho} \mathbf{m}^{-\frac{1}{\rho}}$ denote an auxiliary parameter. Under autarky, $\psi_A = \frac{w_A}{\mathbf{m}} = \left(\frac{R}{\rho}\right)^\rho \frac{1}{\mathbf{m}} = Z^\rho$ is a function of the parameters $\{R, \rho, \mathbf{m}\}$. Figure 15 shows the parameter configuration for multiple steady states under financial integration in the (λ, ψ_A) space and in the $\{\lambda, Z\}$ space, respectively. The dashed line in the left panel shows the threshold value $\tilde{\psi}_A$ defined in lemma 3. According to the proof of propositions 3 and 6, figure 15 is the

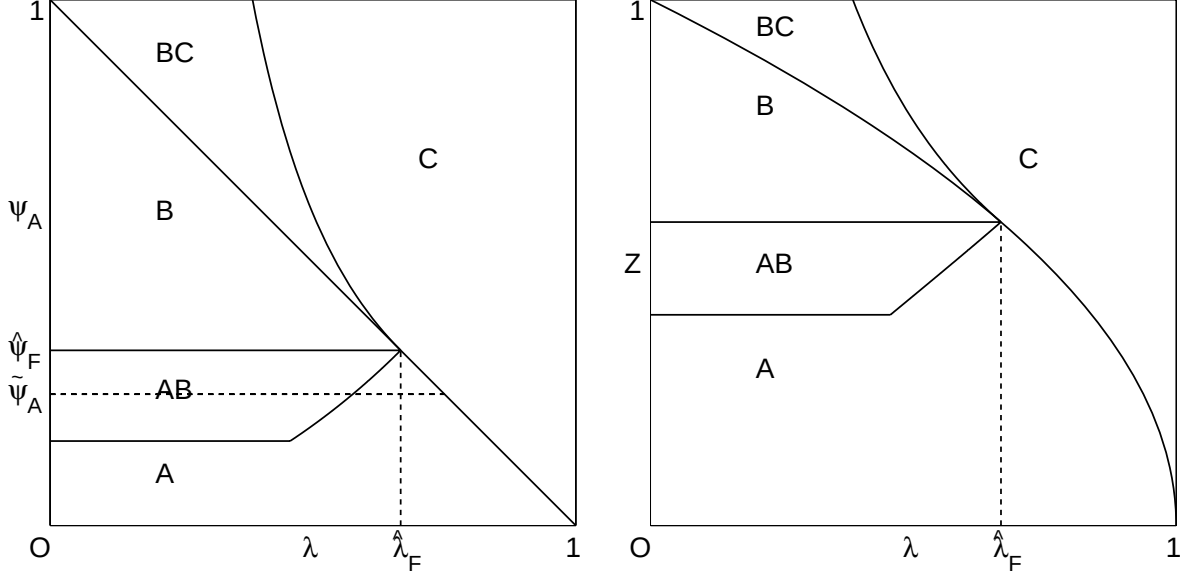


Figure 15: Parameter Configuration for Symmetry Breaking: Setting **F**

limiting case of figure 7 with $\theta \rightarrow 0$.

The right panel of figure 15 is almost identical as figure 5 of Matsuyama (2004) except for the boundary between region AB and A. Implicitly, the mass of entrepreneurs cannot exceed the total mass of population in each generation, $\tau_t^i \leq 1$. Taking that into account, the boundary between region AB and A is characterized by a piecewise function with two subfunctions.³⁰ This result is confirmed in our model with $\sigma = 1$ in section 4. Thus, we correct an error in figure 5 of Matsuyama (2004).

The laws of motion for wage under financial integration versus under autarky in setting **F** are qualitatively identical as those in our model with $\sigma = 1$ in section 4. See figure 16.

To sum up, the model with the FIR in Matsuyama (2004) is essentially the limiting case of our model with $\sigma = 1$ and $\theta \rightarrow 0$.

A.4 Heterogeneity in Financial Development: Setting **N**

In section 5, we introduce the cross-country heterogeneity in financial development and address three issues for a small open economy in the case of $\sigma = 1$. In order to highlight the extensive-margin effect in the presence of the MIR and financial frictions, we conduct the parallel analysis in a two-country version of setting **N**.

By assumption, $\lambda^W < 1 - \tau$ so that the borrowing constraints are binding in country **W** at the autarkic steady state with $r^W = \frac{\lambda^W \rho}{1 - \tau}$.

A.4.1 Threshold Conditions for Financial Integration

The dashed curve in figure 17 shows the law of motion for wage in country **S** under autarky. As in the case of $\sigma = 0$ in subsection 4.1, the negative interest rate response to income changes under autarky ensures that, given $r^* = r^W$, there exists a unique steady state under financial

³⁰See the proof of proposition 6 for the analytical characterization of the two subfunctions.

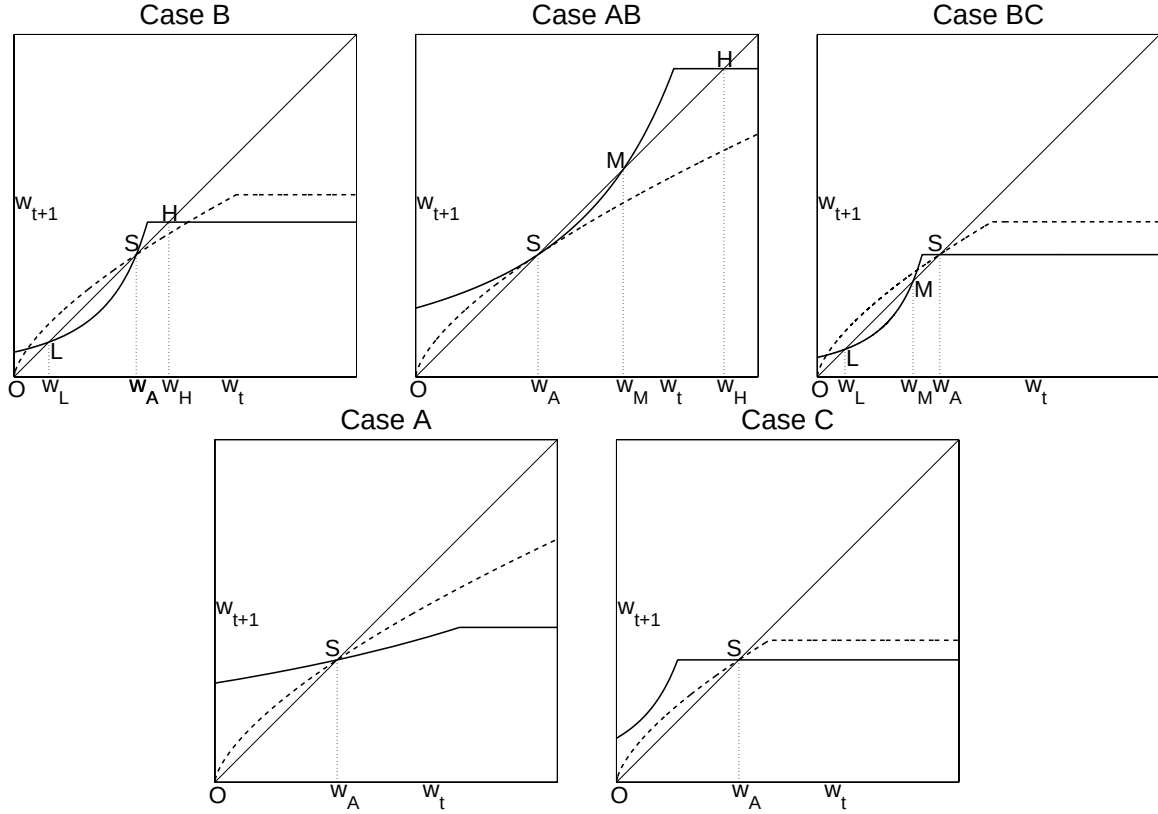


Figure 16: Laws of Motion for Wage: Setting **F**

integration where the wage rate is

$$w^S = \begin{cases} \left[\frac{(1-\tau)\lambda^S}{\lambda^W} + \tau \right]^\rho w_A, & \text{if } \lambda^S \leq \hat{\lambda}^S; \\ \left(\frac{1-\tau}{\lambda^W} \right)^\rho w_A, & \text{if } \lambda^S > \hat{\lambda}^S, \end{cases} \quad \text{where } \hat{\lambda}^S \equiv 1 - \frac{\tau}{1-\tau} \lambda^W > \lambda^W. \quad (\text{A.7})$$

The thick (thin) curve in the left panel of figure 17 shows the law of motion for wage if the level of financial development in country S is significantly (slightly) larger than that in country W, while the right panel shows the case where $\lambda^S < \lambda^W$.

In setting **N**, there is a unique steady state under financial integration and, for $\lambda^S < \hat{\lambda}^S$, the steady-state income effect in country i is roughly log-linear in $\frac{\lambda^S}{\lambda^W}$, regardless of the initial income; in our model with the constant MIR ($\sigma = 1$), financial integration may lead to multiple steady states and the steady-state income effect in country i may depend on the initial income.

A.4.2 Domestic Financial Crises and Capital Controls

With the same parameter values as for the left panel of figure 10, the dashed curve in the left panel of figure 18 shows the law of motion for wage under autarky, while the thin (thick) curve shows that in the scenario of λ_1^S (λ_2^S) under financial integration.

Compared to the case of the constant MIR, the decline in λ^S has the different implications. First, the steady-state income effect is much smaller, according to equation (A.7); second, when λ^S goes back to its initial level, country S returns to its initial steady state, regardless of its initial income. In the absence of the MIR, there is no path dependence and capital controls are not necessary.

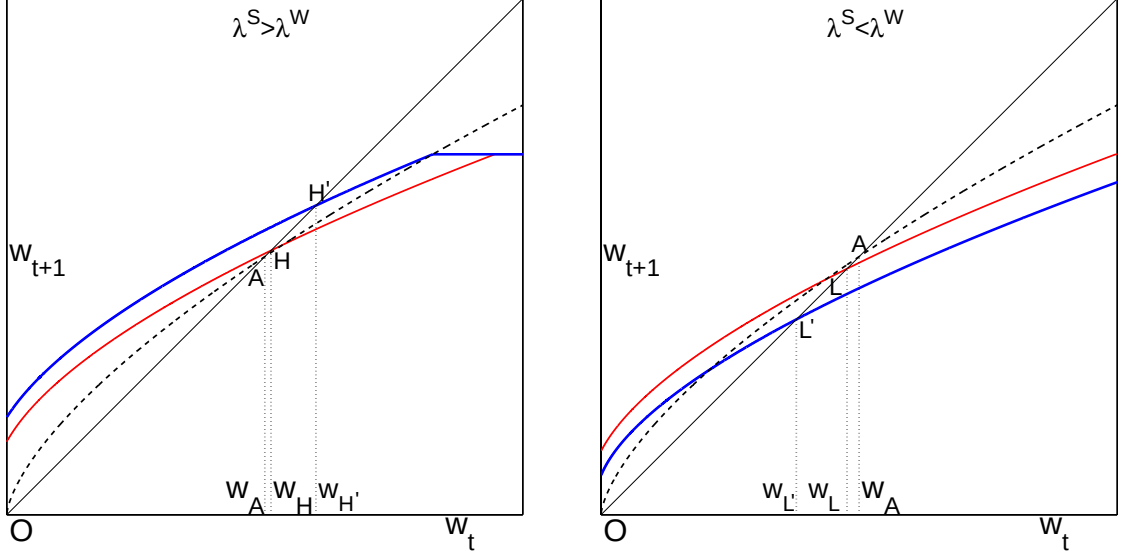


Figure 17: Impacts of Financial Integration: Setting N

A.4.3 World Interest Rate Shocks and Economic Volatility

According to equation (A.7), a decline in λ^W reduces the world interest rate, which affects the steady-state aggregate income in country S roughly in a log-linear way. The steady-state income effects may depend on the level of financial development in country S, which is not as dramatic as in our model with $\sigma = 1$.

B Alternative Specifications

We have made many assumptions to highlight the core mechanism behind symmetry breaking. One can extend our model setting in various aspects and our results should still hold as long as the positive interest rate response to income changes is maintained or reinforced. In this section, we briefly discuss two alternative specifications.³¹

B.1 Endogenizing the Level of Financial Development

The empirical literature has documented extensive evidences on the positive impacts of financial development on economic growth (Buera, Kaboski, and Shin, 2011; Christopoulos and Tsionas, 2004; King and Levine, 1993; Levine, Loayza, and Beck, 2000; Rioja and Valev, 2004). Conversely, economic development also induces the financial sectors to evolve and progress endogenously so as to serve the demand from the real sectors in the form of ameliorating the information asymmetry among market participants as well as improving risk-sharing opportunities (Acemoglu and Zilibotti, 1997; Laeven, Levine, and Michalopoulos, 2015; Song and Thakor, 2010).

For simplicity, we keep the level of financial development exogenous at λ in the analysis. One can relax the assumption and allow the level of financial development to be higher in the rich than in the poor country, $\frac{\partial \lambda_t}{\partial Y_t} > 0$. Given the binding borrowing constraint, combine equations

³¹This section is motivated by the insightful comments of one of our referees.

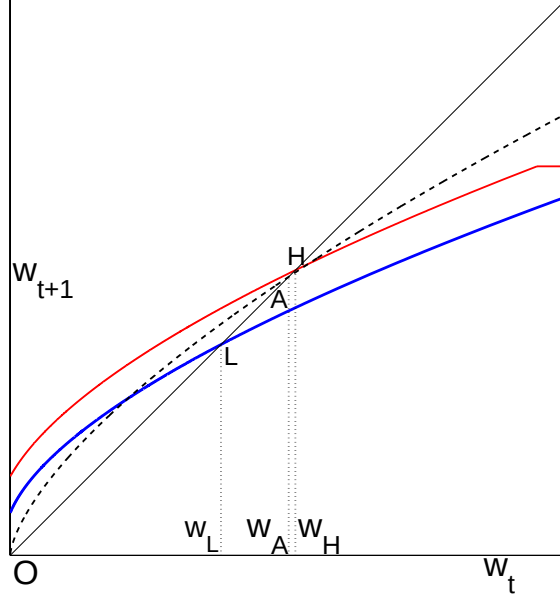


Figure 18: Impacts of Domestic Financial Crisis: Setting N

(21)-(22) to get the autarkic interest rate,

$$\ln r_t = \ln q_{t+1}R + \ln \lambda_t - \ln(1 - \tau_t^{1-\theta}),$$

$$\frac{\partial \ln r_t}{\partial \ln w_t} = \underbrace{\frac{\partial \ln q_{t+1}R}{\partial \ln w_t}}_{\text{the neoclassical effect (-)}} + \underbrace{\frac{\partial \ln \lambda_t}{\partial \ln w_t}}_{\text{the financial-development effect (+)}} - \underbrace{\frac{\partial \ln(1 - \tau_t^{1-\theta})}{\partial \ln \tau_t} \frac{\partial \ln \tau_t}{\partial \ln w_t}}_{\text{the extensive-margin effect}}.$$

- In the case of $\sigma = 1$, the extensive-margin effect is positive $\frac{\partial \ln \tau_t}{\partial \ln w_t} = \sigma > 0$. The positive financial-development effect reinforces the extensive-margin effect, which enhances the positive interest rate response to income changes.
- In the case of $\sigma = 0$, the extensive-margin effect is absent $\frac{\partial \ln \tau_t}{\partial \ln w_t} = 0$. Nevertheless, if the positive financial-development effect can dominate the negative neoclassical effect, the interest rate responds positively to income changes.

In the setting of endogenous financial development, aggregate income affects the interest rate on the aggregate credit demand side; in the setting of the constant MIR, aggregate income affects the interest rate on the aggregate credit supply side. This way, the two settings feature two competing theories on symmetry breaking.

Endogenizing financial development may affect symmetry breaking in two ways.

- If the borrowing constraints are binding, the financial-development effect enhances the positive interest rate response to income change under autarky, which may raise the possibility and the magnitude of symmetry breaking.
- The borrowing constraints are binding iff $\psi_t < 1 - \lambda_t$, where $\frac{\partial \psi_t}{\partial Y_t} > 0$. The financial-development effect makes the borrowing constraints more likely to be slack. In that case, the interest rate declines in aggregate income under autarky, which reduces the possibility and the magnitude of symmetry breaking.

Thus, endogenizing financial development may have ambiguous impacts on the possibility and the magnitude of symmetry breaking.

B.2 Endogenizing the Project Choices

In our model, agents have one indivisible project and those who cannot meet the MIR can only lend out their labor income. Alternatively, one can assume that agents have the two projects, indexed by **M** and **N**. Project **M** has the positive MIR $\mathbf{m}$ and its productivity is R , as in our model with $\sigma = 1$; project **N** has no MIR and its productivity is γR , where $\gamma \in (0, 1)$ denotes the relative productivity.³² Figure 19 shows the production functions of two projects, where $m_{p,t}$ and $k_{p,t+1}$ denote the input and output of project $p \in \{M, N\}$.

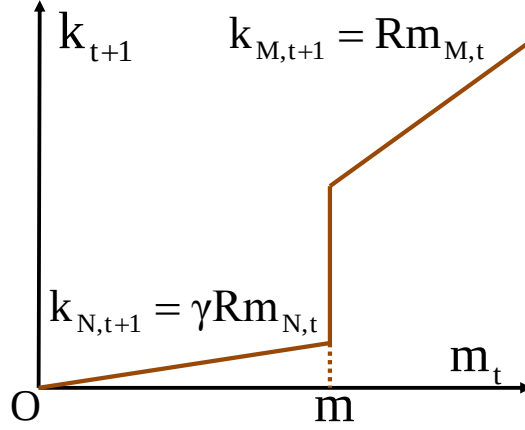


Figure 19: The Production Functions of Two Projects

In the two-project setting, we still call those who can meet the MIR of project **M** as entrepreneurs and the rest as households. Let $\bar{\psi} \equiv 1 - \lambda$ and $\bar{w}_A \equiv \mathbb{F}\bar{\psi}^{\frac{1}{1-\theta}}$. Under autarky, for $w_t > \bar{w}_A$, the borrowing constraints are slack and $r_t = Rq_{t+1} > \gamma Rq_{t+1}$ so that households do not invest in project **N**; for $w_t < \bar{w}_A$, the borrowing constraints are binding, $r_t = \frac{\lambda}{1-\psi_t} Rq_{t+1}$, and whether households invest in project **N** depends on the relative productivity γ and aggregate income.

Let us focus on the interval of $w_t \in (0, \bar{w}_A)$. Figure 20 shows the mass of entrepreneurs τ_t , the equity-investment ratio ψ_t , and the relative interest rate $\Upsilon_t \equiv \frac{r_t}{q_{t+1}R}$ as the functions of $w_t \in (0, \bar{w}_t)$. The solid (dashed) curves show their patterns in the case of $\gamma < \lambda$ ($\gamma > \lambda$).

For $\gamma < \lambda$, $\frac{\lambda}{1-\psi_t} > \lambda > \gamma$ and $r_t = \frac{\lambda}{1-\psi_t} Rq_{t+1} > \gamma Rq_{t+1}$ so that nobody invests in project **N** and the allocation is the same as in our model with $\sigma = 1$. Thus, the solid curves in figure 20 also show the patterns of those variables in our model with $\sigma = 1$.

For $\gamma > \lambda$, let $\underline{\psi} \equiv 1 - \frac{\lambda}{\gamma}$ and $\underline{w}_A \equiv \mathbb{F}\underline{\psi}^{\frac{1}{1-\theta}}$.

- If $w_t \in [\underline{w}_A, \bar{w}_A)$, $\psi_t > \underline{\psi}$ and $r_t = \frac{\lambda}{1-\psi_t} q_{t+1}R > q_{t+1}\gamma R$ so that nobody invests in project **N** and the allocation is the same as in our model with $\sigma = 1$.
- If $w_t \in (0, \underline{w}_A)$, $\psi_t = \underline{\psi}$ and $r_t = \frac{\lambda}{1-\psi_t} Rq_{t+1} = \gamma Rq_{t+1}$ so that households are indifferent between investing in project **N** and lending. The total investments in project **N** and in project **M** are $\int_{\epsilon_t}^{\infty} \frac{(1-\theta)\epsilon_j w_t}{\underline{\psi}} dG(\epsilon_j) = \frac{\epsilon_t}{\underline{\psi}} w_t$ and $(1 - \frac{\epsilon_t}{\underline{\psi}})w_t$ respectively. The

³²Our model can be considered as a special case in this setting with $\gamma = 0$. In a more general setting, one can assume that agents have the access to $N+1$ projects. Let $\mathbf{m}_n$ and R_n denote the MIR and the productivity of project $n = 0, 1, 2, \dots, N$. Projects are ranked according to the MIR ($\mathbf{m}_0 = 0$ and $\mathbf{m}_n > \mathbf{m}_{n-1}$ for $n = 1, 2, \dots, N$) and the productivity ($R_0 \geq 0$ and $R_n > R_{n-1}$ for $n = 1, 2, \dots, N$).

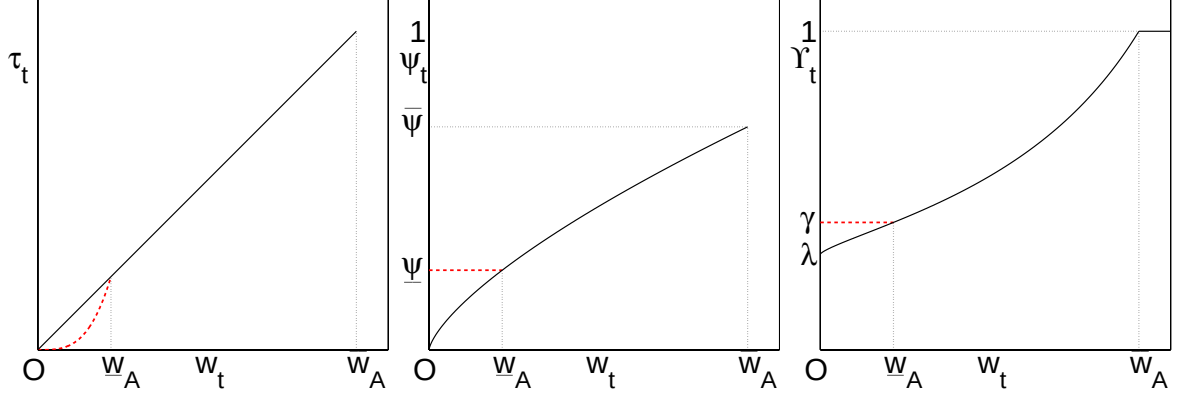


Figure 20: Patterns of Endogenous Variables: $w_t \in (0, \bar{w}_A)$

aggregate output of capital goods in period $t + 1$ is $K_{t+1} = \Gamma_t R w_t$, where $\Gamma_t \equiv \gamma + (1 - \gamma)^{\frac{-\frac{1-\theta}{\psi}}{\psi}}$ measures the allocation efficiency.³³

In our one-project model, aggregate saving is entirely used to finance the productive project under autarky, thanks to the "unconstrained" adjustment of the interest rate. In the two-project model, project **N** serves as an outside option for lenders and its marginal rate of return essentially becomes a "floor" for the interest rate. For $\gamma > \lambda$ and $w_t \in (0, \underline{w}_A)$, a fraction of aggregate saving is allocated inefficiently in the less productive project, due to the "interest rate floor". In this case, the higher the aggregate income, the larger the mass of agents who overcome the MIR and invest in the more productive projects, the more efficient the aggregate investment, the lower the price of capital goods and the rate of return to the less productive project, the larger the households' lending. The aggregate credit supply declines on the extensive margin and rises on the intensive margin. As long as project **N** is still invested, the intensive-margin effect dominates so

³³In our model, only project **M** is available so that aggregate saving is fully used to produce capital goods at the rate of R and financial frictions only distort the interest rates, $r_t < q_{t+1}R < \Omega_t$. In the two-project model, if $\gamma > \lambda$ and $w_t \in (0, \underline{w}_A)$, a fraction of domestic saving is allocated in project **N** and the mass of agents investing in project **M** is lower than in the one-project model. In this case, aggregate allocation is inefficient $\Gamma_t < 1$; according to equation (9), $\epsilon_t = \frac{\psi F}{w_t}$ and $\frac{\partial \Gamma_t}{\partial w_t} > 0$. The higher the aggregate income, the larger the mass of agents who meet the MIR, the larger the fraction of aggregate saving allocated in project **M**, the more efficient the aggregate allocation. Besides the interest rate, financial frictions also distort aggregate efficiency through resource allocation in the two-project model.

Matsuyama (2007) develops a multi-project model where projects differ in terms of productivity and pledgeability. With a sufficiently low net wealth, the credit goes to the project with the highest pledgeable value rather than to the most productive project; with a sufficiently high net wealth, the borrowing constraints are slack so that the credit goes to the most productive project. Matsuyama (2007) shows that an endogenous shift in the composition of the credit across projects with different productivity levels may lead to multiple steady states, which helps explain a variety of economic phenomena, e.g., credit traps, endogenous credit cycles, credit collapse, growth miracles, leapfrogging, and etc.

In our two-project model, if $\gamma > \lambda$ and $w_t \in (0, \underline{w}_t)$, both projects are invested in equilibrium and the endogenous allocation of aggregate saving among the two projects affects the aggregate efficiency. Thus, our model shares the same feature as Matsuyama (2007) in the sense that, in the presence of financial frictions, aggregate efficiency is endogenously determined by the resource allocation among projects with different productivity. This asset allocation effect competes with the decreasing MPK, which may lead to multiple steady states under autarky. The detailed analysis is beyond the scope of this paper.

that the interest rate falls in aggregate income.

Lemma 4. *If $\gamma < \lambda[1 + (1 - \theta)(1 - \alpha)]$, lemma 2 still holds and so does proposition 3.*

- According to Lemma 4, if the productivity difference of the two project is small, i.e., $\gamma > \lambda[1 + (1 - \theta)(1 - \alpha)]$, allowing the endogenous project choices affects the interest rate response to income changes, which may reduce the possibility of symmetry breaking.
- If the productivity difference of the two project is large $\gamma < \lambda[1 + (1 - \theta)(1 - \alpha)]$, the condition specified in lemma 2 still holds so that the symmetry breaking results are unaffected. Thus, when symmetry breaking happens, the endogenous project choice and resource allocation may amplify the size of income polarization.

C Proofs

Proof of Proposition 1

Proof. According to equation (16), the law of motion for wage is log-linear, $\ln w_{t+1} = \alpha \ln w_t + \alpha \ln \frac{R}{\rho}$, with a slope less than unity, $\alpha < 1$. Thus, there exists a unique and stable steady state with $w_A = \left(\frac{R}{\rho}\right)^\rho$. $\square$

Proof of Lemma 1

Proof. The proof consists of two steps.

Step 1: the interest rate under the binding borrowing constraints

Combine equations (1)-(2) to get the factor price equation,

$$q_t^\alpha w_t^{1-\alpha} = 1. \quad (\text{C.1})$$

Rewrite the binding borrowing constraints (6) into $r_t = \frac{\lambda}{1-\psi_t} q_{t+1} R$ and combine it with equations (16) and (C.1) to get equation (17).

According to the credit market clearing equation,

$$D_t = w_t \left(\frac{1}{\psi_t} - 1 \right) \underline{\epsilon}_t^{-(\frac{1}{\theta}-1)}, \quad S_t = w_t [1 - \underline{\epsilon}_t^{-(\frac{1}{\theta}-1)}], \quad D_t = S_t, \Rightarrow \psi_t = \underline{\epsilon}_t^{-(\frac{1}{\theta}-1)}. \quad (\text{C.2})$$

Combine equations (9) and (C.2) to get equation (18).

Thus, one can use equations (17)-(18) to solve ψ_t and r_t as the functions of w_t .

Step 2: the condition for the binding borrowing constraints

According to equation (17), the borrowing constraints are binding iff $r_t < q_{t+1} R$ or equivalently $\psi_t \in (0, \bar{\psi})$, where $\bar{\psi} \equiv 1 - \lambda$.

Combine $w_t = (1 - \alpha)Y_t$ with equation (18) to get ψ_t as a function of Y_t ,

$$\sigma \ln Y_t = \frac{1}{1 - \theta} \ln \psi_t + \ln \mathbf{m} - \ln(1 - \theta)(1 - \alpha), \quad (\text{C.3})$$

$$\frac{\partial \ln \psi_t}{\partial \ln \mathbf{m}} = -(1 - \theta) < 0, \quad \frac{\partial \ln \psi_t}{\partial \ln Y_t} = \sigma(1 - \theta), \Rightarrow \text{sgn} \left(\frac{\partial \psi_t}{\partial Y_t} \right) = \text{sgn}(\sigma). \quad (\text{C.4})$$

In the boundary case where the borrowing constraints are weakly binding, $\psi_t = \bar{\psi}$, equation (C.3) can be rewritten as $\mathbf{m} = (Y_t)^\sigma \Lambda$.

Taking into account equations (C.4), for $\sigma = 0$ and $\sigma = 1$, the condition for the binding borrowing constraints $\psi_t < \bar{\psi}$ can be restated as $\mathbf{m} > (Y_t)^\sigma \Lambda$ and the condition for the slack borrowing constraints $\psi_t > \bar{\psi}$ can be restated as $\mathbf{m} < (Y_t)^\sigma \Lambda$. $\square$

Proof of Lemma 2

Proof. If the borrowing constraints are slack, $\frac{\partial \ln r_t}{\partial \ln w_t} = -(1 - \alpha) < 0$. See equation (20).

If the borrowing constraints are binding, $\psi_t \in (0, \bar{\psi})$ and rewrite equation (17) as

$$\ln r_t = (\alpha - 1) \ln w_t - \ln(1 - \psi_t) + \ln \lambda R^\alpha \rho^{1-\alpha}, \quad (\text{C.5})$$

$$\Rightarrow \frac{\partial \ln r_t}{\partial \ln w_t} = \underbrace{\alpha - 1}_{\text{the neoclassical effect}} + \underbrace{\frac{\sigma(1 - \theta)}{\frac{1}{\psi_t} - 1}}_{\text{the extensive-margin effect}}. \quad (\text{C.6})$$

For $\sigma = 0$, according to equation (C.6), the extensive-margin effect is absent and, due to the neoclassical effect, $\frac{\partial \ln r_t}{\partial \ln w_t} = \alpha - 1 < 0$.

For $\sigma = 1$, according to equation (C.6), the extensive-margin effect works against the neoclassical effect. For $\psi_t \in (0, \tilde{\psi}_A)$, the neoclassical effect dominates so that $\frac{\partial \ln r_t}{\partial \ln w_t} < 0$; for $\psi_t \in (\tilde{\psi}_A, \bar{\psi})$, the extensive-margin effect dominates so that $\frac{\partial \ln r_t}{\partial \ln w_t} > 0$. $\square$

Proof of Proposition 3

Proof. The proof consists of three steps. For simplicity, we suppress the country index i .

Step 1: the law of motion for wage under financial integration

As a small open economy, country i takes the world interest rate as exogenously given at r^* . If $Rq_{t+1} > r^*$, the borrowing constraints are binding, which gives equation (28). In this case, the law of motion for wage depends on the mass-of-entrepreneurs constraints.

- If the mass-of-entrepreneurs constraint is slack, $\tau_t < 1$, not all agents can meet the MIR $\underline{\epsilon}_t = \tau_t^{-\theta} > 1$, and equation (9) becomes $\underline{\epsilon}_t^i = \frac{\psi_t^i \mathbb{F}}{(w_t^i)^\sigma}$. Combine aggregate investment $\frac{K_{t+1}}{R} = \int_{\underline{\epsilon}_t}^\infty \frac{l_j w_t}{\psi_t} dG(l_j) = \frac{w_t \underline{\epsilon}_t^{-(\frac{1}{\theta}-1)}}{\psi_t}$ with equations (1)-(2) and (9) to get equation (29). Use equations (28) and (29) to get

$$\frac{\partial w_{t+1}}{\partial w_t} = \frac{w_{t+1}}{w_t} \frac{\alpha[\sigma(1 - \theta) + \theta]}{\mathbb{K}} > 0, \text{ where } \mathbb{K} \equiv (1 - \alpha) \frac{1 - \psi_t}{\psi_t} + \theta > \theta. \quad (\text{C.7})$$

- If the mass-of-entrepreneurs constraint is binding, $\tau_t = 1$, all agents can meet the MIR $\underline{\epsilon}_t = \tau_t^{-\theta} = 1$, and equation (9) becomes $\underline{\epsilon}_t^i = 1$. Combine aggregate investment $\frac{K_{t+1}}{R} = \frac{w_t}{\psi_t}$ with equations (1)-(2) to get (30). Use equations (28) and (30) to get

$$\frac{\partial w_{t+1}}{\partial w_t} = \frac{w_{t+1}}{w_t} \frac{1}{1 + \frac{1}{\psi_t \rho}} > 0. \quad (\text{C.8})$$

According to equations (C.7)-(C.8), the law of motion for wage is monotonically increasing when the borrowing constraints are binding.

The borrowing constraints are slack, iff $Rq_{t+1} = r^*$. Combine it with equation (C.1) to get $w_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$. In this case, the law of motion for wage is flat at $w_{t+1} = \bar{w}_{t+1} \equiv w_A \left(\frac{\rho}{r^*}\right)^\rho$, which is also the upper bound of w_{t+1} .

Step 2: the threshold condition for two constraints

Here, we derive the condition under which the mass-of-entrepreneurs constraint and the borrowing constraints are binding and then describe the law of motion for wage.

Combine $\tau_t = \underline{\epsilon}_t^{-\frac{1}{\theta}}$ with equations (9) and (28)-(30) to get

$$\text{for } \tau_t = 1, \quad \frac{\partial \ln \psi_t}{\partial \ln w_t} = \frac{1}{1 + \frac{\psi_t}{1-\psi_t} \frac{1}{1-\alpha}} \in (0, 1); \quad (\text{C.9})$$

$$\text{for } \tau_t < 1, \quad \frac{\partial \ln \psi_t}{\partial \ln w_t} = [\sigma(1-\theta) + \theta] \left(1 - \frac{\theta}{\mathbb{K}}\right) > 0, \quad \text{and} \quad (\text{C.10})$$

$$\frac{\partial \ln \tau_t}{\partial \ln w_t} = \frac{1}{\theta} \left[\sigma - \frac{\partial \ln \psi_t}{\partial \ln w_t} \right] = \begin{cases} -\frac{(\mathbb{K}-\theta)}{\mathbb{K}} < 0, & \text{if } \sigma = 0; \\ \frac{1}{\mathbb{K}} > 0, & \text{if } \sigma = 1. \end{cases} \quad (\text{C.11})$$

Thus, ψ_t increases in w_t . For $w_t \rightarrow 0$, $\psi_t \rightarrow 0$ and $w_{t+1} \rightarrow \lambda^\rho \bar{w}_{t+1}$, according to equation (28). Thus, the law of motion for wage has a positive vertical intercept at $\underline{w}_{t+1} \equiv \lambda^\rho \bar{w}_{t+1}$, which is also the lower bound of w_{t+1} .

In the following, we first use ψ_t to characterize the threshold condition of $\tau_t = 1$ (or equivalently $\underline{\epsilon}_t = 1$) and then compute the corresponding value of w_t .

In the case of $\sigma = 0$, put $\underline{\epsilon}_t = 1$ into equation (9) to get a threshold value $\psi_0 \equiv \frac{1}{\mathbb{F}}$.

- If $\psi_0 > \bar{\psi}$ or equivalently $\frac{1}{\mathbb{F}} > 1 - \lambda$, the law of motion for wage is defined over two intervals. Let $\bar{w}_F = (1 - \lambda)w_A \left(\frac{\rho}{r^*}\right)^{\frac{1}{1-\alpha}}$.
 - For $w_t \in (0, \bar{w}_F)$, $\tau_t = 1$ and the borrowing constraints are binding, $\psi_t \in (0, \bar{\psi})$. The law of motion for wage is defined by equations (28)-(29).
 - For $w_t > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1} = r^*$, and the law of motion for wage is flat at $w_{t+1} = \bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$.
- If $\psi_0 < \bar{\psi}$ or equivalently $\frac{1}{\mathbb{F}} < 1 - \lambda$, the law of motion for wage is defined over three intervals. Let $\hat{w}_F \equiv \frac{1}{\mathbb{F}} \left[\frac{\lambda}{1-\frac{1}{\mathbb{F}}} \right]^{\frac{1}{1-\alpha}} w_A \left(\frac{\rho}{r^*}\right)^{\frac{1}{1-\alpha}}$ and $\bar{w}_F \equiv \frac{[(1-\lambda)\mathbb{F}]^{\frac{1}{\theta}}}{\mathbb{F}} w_A \left(\frac{\rho}{r^*}\right)^{\frac{1}{1-\alpha}}$.
 - For $w_t \in (0, \hat{w}_F)$, $\tau_t = 1$ and the borrowing constraints are binding, $\psi_t \in (0, \psi_0)$. The law of motion for wage is defined by equations (28)-(29).
 - For $w_t \in (\hat{w}_F, \bar{w}_F)$, $\tau_t < 1$ and the borrowing constraints are binding, $\psi_t \in (\psi_0, \bar{\psi})$. The law of motion for wage is defined by equations (28) and (30).
 - For $w_t > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1} = r^*$, and the law of motion for wage is flat at $w_{t+1} = \bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$.

In the case of $\sigma = 1$, put $\underline{\epsilon}_t = 1$ into equation (9) and combine it with equations (28) and (29) to get a threshold value $\psi_1 = 1 - \frac{\lambda\rho}{r^*} \left[\frac{w_A}{\mathbb{F}} \right]^{1-\alpha}$.

- if $\psi_1 > \bar{\psi}$ or equivalently $\rho \left[\frac{w_A}{\mathbb{F}} \right]^{1-\alpha} < r^*$, the law of motion for wage is defined over two intervals. Let $\bar{w}_F \equiv \frac{1-\lambda}{\psi_A} w_A \left(\frac{\rho}{r^*}\right)^{\frac{\theta}{1-\alpha}}$.
 - For $w_t \in (0, \bar{w}_F)$, $\tau_t < 1$ and the borrowing constraints are binding, $\psi_t \in (0, \bar{\psi})$. The law of motion for wage is defined by equations (28) and (29).
 - For $w_t > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1} = r^*$, and the law of motion for wage is flat at $w_{t+1} = \bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$.
- if $\psi_1 < \bar{\psi}$ or equivalently $\rho \left[\frac{w_A}{\mathbb{F}} \right]^{1-\alpha} > r^*$, the law of motion for wage is defined over three intervals. Let $\hat{w}_F \equiv \mathbb{F}\psi_1$ and $\bar{w}_F \equiv (1 - \lambda)w_A \left(\frac{\rho}{r^*}\right)^{\frac{1}{1-\alpha}}$.

- For $w_t \in (0, \hat{w}_F)$, $\tau_t < 1$ and the borrowing constraints are binding, $\psi_t \in (0, \psi_1)$. The law of motion for wage is defined by equations (28) and (29).
- For $w_t \in (\hat{w}_F, \bar{w}_F)$, $\tau_t = 1$ and the borrowing constraints are binding, $\psi_t \in (\psi_1, \bar{\psi})$. The law of motion for wage is defined by equations (28) and (30).
- For $w_t > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1} = r^*$, and the law of motion for wage is flat at $w_{t+1} = \bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*} \right)^\rho$.

Step 3: the conditions for multiple steady states under financial integration

As proved above, for $w_t \in (0, \infty)$, w_{t+1} is bounded between $\underline{w}_{t+1} > 0$ and $\bar{w}_{t+1} < \infty$; according to equations (C.8) and (C.7), the law of motion for wage is upward-sloping (flat) under financial integration, if the borrowing constraints are binding (slack). Thus, the law of motion for wage must cross the 45° line from the left at least once and hence, there exists at least a stable steady state under financial integration.

Multiple steady states may arise under financial integration, iff the slope of the law of motion for wage is equal to or larger than unity at one steady state.

For $\sigma = 0$, use equations (C.8) and (C.7) to evaluate $\frac{\partial w_{t+1}}{\partial w_t}$ at any steady state(s),

$$\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_{t+1}=w_t} = \begin{cases} \frac{1}{1+\frac{1}{\psi_t \rho}} \in (0, 1) & \text{and } \tau = 1, \text{ if } w_t \in (0, \min\{\hat{w}_F, \bar{w}_F\}); \\ \frac{1}{1+\frac{1}{\psi_t \rho} + (\frac{1}{\theta}-1)\frac{1-\psi_t}{\rho \psi_t}} \in (0, 1) & \text{and } \tau < 1, \text{ if } w_t \in (\min\{\hat{w}_F, \bar{w}_F\}, \bar{w}_F]; \\ 0, & \text{if } w_t > \bar{w}_F. \end{cases} \quad (\text{C.12})$$

Since the slope of the law of motion for wage at any steady state is non-negative and less than unity, the steady state under financial integration must be unique and stable. See figure 5.

For $\sigma = 1$, use equations (C.8) and (C.7) to evaluate $\frac{\partial w_{t+1}}{\partial w_t}$ at the steady state(s),

$$\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_{t+1}=w_t} = \begin{cases} 1 + \left(\psi_t - \frac{1-\alpha}{1-\theta} \right) \frac{(1-\theta)}{\mathbb{K} \psi_t} & \text{and } \tau < 1, \text{ if } w_t \in (0, \min\{\hat{w}_F, \bar{w}_F\}); \\ \frac{1}{1+\frac{1}{\psi_t \rho}} \in (0, 1) & \text{and } \tau = 1, \text{ if } w_t \in (\min\{\hat{w}_F, \bar{w}_F\}, \bar{w}_F]; \\ 0, & \text{if } w_t > \bar{w}_F. \end{cases} \quad (\text{C.13})$$

- The slope of the law of motion for wage is non-negative and less than unity at the steady state in the interval of $w_t > (\min\{\hat{w}_F, \bar{w}_F\}, \bar{w}_F)$.
- The slope of the law of motion for wage is positive and can exceed unity at the steady state in the interval of $w_t \in (0, \min\{\hat{w}_F, \bar{w}_F\}]$, if $\psi_t > \frac{1-\alpha}{1-\theta}$. Here, $\{\alpha, \theta\}$ are critical for the existence of multiple steady states under financial integration.
 - If $\theta > \alpha$, $\frac{1-\alpha}{1-\theta} > 1 > \bar{\psi} > \psi_t$ so that $\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_{t+1}=w_t} \in [0, 1)$ for $w_t \in (0, \infty)$. Thus, the steady state is unique and stable under financial integration.
 - If $\theta < \alpha$, multiple steady state may arise under financial integration, depending on the model parameters.

In the following, we focus on the case of $\theta < \alpha$. Use equations (28)-(30) to get

$$\frac{\partial w_{t+1}}{\partial w_t^2} = \begin{cases} \left[\frac{1-2\theta}{1-\theta} - \psi_t \frac{1-\alpha-\theta}{1-\alpha} \right] \frac{\mathbb{J}(1-\alpha)^2(1-\theta)^2}{w_t \psi_t \mathbb{K}^3}, & \text{if } \tau_t < 1; \\ -\frac{\psi_t}{(1-\alpha)(1+\psi_t \rho)^2} \frac{\mathbb{J}}{w_t} < 0, & \text{if } \tau_t = 1. \end{cases} \quad (\text{C.14})$$

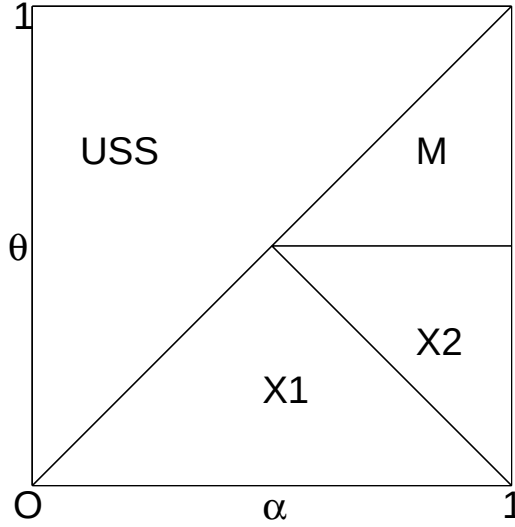


Figure 21: The Shape of the Law of Motion for Wage: $\sigma = 1$ and $\tau_t > 1$

For $w_t < \bar{w}_F$, the law of motion for wage is concave in the case of $\tau_t = 1$, while its shape depends on the parameters $\{\theta, \alpha, \lambda\}$ in the case of $\tau_t < 1$.

Figure 21 shows the shape of the law of motion for wage in four scenarios in the $\{\alpha, \theta\}$ space. Let $\check{\psi}_F \equiv 1 + \frac{\theta(\alpha-\theta)}{(1-\alpha-\theta)(1-\theta)}$.

- For $\{\theta, \alpha\}$ in region USS, $\theta > \alpha$ and, as proved above, there exists a unique, stable steady state, regardless of the shape of the law of motion for wage.
- For $\{\theta, \alpha\}$ in region X1, $\alpha > \theta$ and $1 - \alpha - \theta > 0$ so that $\check{\psi}_F > 1 > \bar{\psi} > \psi_t$ and $\frac{1-2\theta}{1-\theta} - \psi_t \frac{1-\alpha-\theta}{1-\alpha} = (\check{\psi}_F - \psi_t) \frac{1-\alpha-\theta}{1-\alpha} > 0$. Thus, $\mathbb{H} > 0$ and the law of motion for wage is convex for $\psi_t \in (0, \bar{\psi})$.
- For $\{\theta, \alpha\}$ in region X2, $1 - 2\theta > 0$ and $1 - \theta - \alpha < 0$ so that $\frac{1-2\theta}{1-\theta} > 0 > \psi_t \frac{1-\alpha-\theta}{1-\alpha}$. Thus, $\mathbb{H} > 0$ and the law of motion for wage is convex for $\psi_t \in (0, \bar{\psi})$.
- For $\{\theta, \alpha\}$ in region M, $\alpha > \theta$ and $1 - \theta - \alpha < 0$ so that $\check{\psi}_F = 1 - \frac{\theta(\alpha-\theta)}{(\alpha+\theta-1)(1-\theta)} < 1$.
 - If $\frac{\theta(\alpha-\theta)}{(\alpha+\theta-1)(1-\theta)} < \lambda$ or equivalently $\check{\psi}_F > \bar{\psi} > \psi_t$, $\frac{1-2\theta}{1-\theta} - \psi_t \frac{1-\alpha-\theta}{1-\alpha} = (\check{\psi}_F - \psi_t) \frac{1-\alpha-\theta}{1-\alpha} < 0$. Thus, $\mathbb{H} < 0$ and the law of motion for wage is concave for $\psi_t \in (0, \bar{\psi})$.
 - If $\frac{\theta(\alpha-\theta)}{(\alpha+\theta-1)(1-\theta)} > \lambda$ or equivalently $\check{\psi}_F < \bar{\psi}$. Then, for $\psi_t \in (0, \check{\psi}_F)$, $\mathbb{H} < 0$ and the law of motion for wage is concave; for $\psi_t \in (\check{\psi}_F, \bar{\psi})$, $\mathbb{H} > 0$ and the law of motion for wage is convex.

In the following, we analyze three cases where the multiple steady states arise in the case of $\sigma = 1$ and $r^* = r_A$, as shown in the upper-panels of figure 8. We also derive the boundary conditions for the five regions of figure 7.

Case 1: consider the upper-right triangle of figure 7 where the borrowing constraints are slack at the autarkic steady state with $r_A = \rho$ and $\psi_A \in (\bar{\psi}, 1]$. Given $r^* = r_A = \rho$, $\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_A} = 0$ so that the autarkic steady state is still stable under financial integration. Compare the upper-right and the lower-right panels of figure 8. Multiple steady states arise if the law of motion for wage crosses the 45° line at point M with $\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_M} \geq 1$ and $w_M < \bar{w}_F$ or equivalently $\psi_M < \bar{\psi}$.

Use equation (C.13) to get $\psi_M > \frac{1-\alpha}{1-\theta}$. Combine it with (9) and (28)-(29). Then, region BC in figure 7 is defined by,

$$\lambda < \frac{\alpha - \theta}{1 - \theta} \text{ and } \bar{\psi} < \psi_A \leq \min\{\psi_{BC}, 1\}, \text{ where } \psi_{BC} \equiv \frac{1 - \alpha}{1 - \theta} \left[\frac{\alpha - \theta}{(1 - \theta)\lambda} \right]^{\frac{\alpha - \theta}{1 - \alpha}}. \quad (\text{C.15})$$

Case 2: consider the lower-left triangle of figure 7 where the borrowing constraints are binding at the autarkic steady state with $\psi_A \in (0, \bar{\psi})$ and $r_A = \frac{\lambda\rho}{1-\psi_A} < \rho$. As shown in the upper-left panel of figure 8, given $r^* = r_A$ under financial integration, case B arises iff $\frac{\partial w_{t+1}}{\partial w_t} \big|_{w_A} \geq 1$ at the autarkic steady state. Use equation (C.13) to get

$$\lambda < \frac{\alpha - \theta}{1 - \theta} \text{ and } \hat{\psi}_F < \psi_A \leq \bar{\psi}, \text{ where } \hat{\psi}_F \equiv \frac{1 - \alpha}{1 - \theta}. \quad (\text{C.16})$$

which defines region B in figure 7.

Case 3: consider the region with $\psi_A < \min\{\hat{\psi}_F, \bar{\psi}\}$ in figure 7. Since $\frac{\partial w_{t+1}}{\partial w_t} \big|_{w_A} < 1$, the autarkic steady state is still stable under financial integration.

Given $\psi_A < \min\{\hat{\psi}_F, \bar{\psi}\}$, if the law of motion for wage is tangent with the 45° line at point M, $\psi_M = \frac{1-\alpha}{1-\theta}$ must hold, according to equation (C.13). Combine it with $w_{t+1} = w_t$ and equations (28) and (29) to get $\psi_A(1 - \psi_A)^{\frac{\alpha-\theta}{1-\alpha}} = \psi_M(1 - \psi_M)^{\frac{\alpha-\theta}{1-\alpha}}$ and the unique solution for ψ_A is $\psi_A = \psi_M = \frac{1-\alpha}{1-\theta}$, which contradicts the assumption $\psi_A < \min\{\hat{\psi}_F, \bar{\psi}\}$. Thus, the law of motion cannot be tangent with the 45° line in this case.

According to step 2, the law of motion for wage has one or two kinks, depending on the relative size of ψ_1 and $\bar{\psi}$. Compare the upper-middle and the lower-left panel of figure 8. Financial integration may lead to multiple steady states in two subcases.

- Case 3.1: as shown in step 2, given $r^* = r_A = \frac{\lambda\rho}{1-\psi_A} < \rho$, for $\psi_1 > \bar{\psi}$ or equivalently $(1 - \psi_A)\psi_A^{\frac{1-\alpha}{1-\theta}} < \lambda$, the law of motion for wage has a kink at $w_t = \bar{w}_t$. Multiple steady states arise if the kink is above the 45° line, $\bar{w}_{t+1} \geq \bar{w}_F$, which gives

$$\psi_A(1 - \psi_A)^{\frac{\alpha-\theta}{1-\alpha}} \geq (1 - \lambda)\lambda^{\frac{\alpha-\theta}{1-\alpha}}, \psi_A < \min\{\hat{\psi}_F, \bar{\psi}\}, \text{ and } (1 - \psi_A)\psi_A^{\frac{1-\alpha}{1-\theta}} < \lambda. \quad (\text{C.17})$$

- Case 3.2: for $\psi_1 < \bar{\psi}$, or equivalently $(1 - \psi_A)\psi_A^{\frac{1-\alpha}{1-\theta}} > \lambda$, the law of motion for wage has two kinks at $w_t = \hat{w}_F$ and $w_t = \bar{w}_F$. Multiple steady states arise if the kink at $w_t = \hat{w}_F$ is above the 45° line, which gives

$$(2 - \psi_A)\psi_A^{\frac{1-\alpha}{1-\theta}} \geq 1, \psi_A < \min\{\hat{\psi}_F, \bar{\psi}\}, \text{ and } (1 - \psi_A)\psi_A^{\frac{1-\alpha}{1-\theta}} > \lambda. \quad (\text{C.18})$$

Region AB is defined by conditions (C.17) and (C.18). □

Proof of Corollary 1

Proof. The proof follows closely that for lemma 1. □

Proof of Proposition 4

Proof. According to figure 9, given ψ_A and $r^* = \frac{\lambda^W \rho}{1 - \psi_A}$ where $\lambda^W < 1 - \psi_A$, we compute $\bar{\lambda}^S$ by making the law of motion for wage tangent with the 45° line at point M in the convex part, $\psi_M = \frac{1-\alpha}{1-\theta} < 1 - \bar{\lambda}^S$. Following the proof of proposition 3, we get

$$\bar{\lambda}^S \equiv \lambda^W \frac{(1 - \psi_M) \psi_M^{\frac{1-\alpha}{\alpha-\theta}}}{(1 - \psi_A) \psi_A^{\frac{1-\alpha}{\alpha-\theta}}} > \lambda^W. \quad (\text{C.19})$$

For $\lambda^S > \bar{\lambda}^S$, given ψ_A and $\lambda^W < 1 - \psi_A$, the law of motion for wage crosses the 45° line once and only once at the concave part or at the flat part where the steady-state wage rate is higher than w_A . See step 2 in the proof of proposition 3.

According to figure 9, given ψ_A and $r^* = \frac{\lambda^W \rho}{1 - \psi_A}$ where $\lambda^W < 1 - \psi_A$, we compute $\underline{\lambda}^S$ by making the kink point of the law of motion for wage on the 45° line. According to step 2 in the proof of proposition 3, let $\psi_1 = 1 - \frac{\lambda^S \rho}{r^*} \psi_A^{\frac{1-\alpha}{1-\theta}}$.

- If $\psi_1 > 1 - \lambda^S$ or equivalently $\lambda^W > (1 - \psi_A) \psi_A^{\frac{1-\alpha}{1-\theta}}$, the law of motion for wage is defined over two intervals with the kink point at $w_t = \bar{w}_F \equiv \frac{1-\lambda}{\psi_A} w_A \left(\frac{\rho}{r^*}\right)^{\frac{\theta}{1-\alpha}}$ and $w_{t+1} = \bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$. By setting $\bar{w}_{t+1} = \bar{w}_F$, we get the threshold value

$$\underline{\lambda}^S \equiv 1 - \psi_A \left(\frac{1 - \psi_A}{\lambda^W} \right)^{\frac{\alpha-\theta}{1-\alpha}}. \quad (\text{C.20})$$

- If $\psi_1 < 1 - \lambda^S$ or equivalently $\lambda^W < (1 - \psi_A) \psi_A^{\frac{1-\alpha}{1-\theta}}$, the law of motion for wage is defined over three intervals and the kink point between the convex and the concave parts is defined as $w_t = \hat{w}_F \equiv \mathbb{F} \psi_1$ and $w_{t+1} = \hat{w}_{t+1} \equiv w_A \left[\frac{\lambda^S (1 - \psi_A)}{(1 - \psi_1) \lambda^W} \right]^\rho$. By setting $\hat{w}_{t+1} = \hat{w}_F$, we get the threshold value

$$\underline{\lambda}^S \equiv \frac{\lambda^W}{1 - \psi_A} \left(\psi_A^{-\frac{1-\alpha}{1-\theta}} - 1 \right). \quad (\text{C.21})$$

For $\lambda^S < \underline{\lambda}^S$, given ψ_A and $\lambda^W < 1 - \psi_A$, the law of motion for wage crosses the 45° line once and only once at the convex part where the steady-state wage rate is lower than w_A .

For the intermediate case where $\lambda^S \in (\underline{\lambda}^S, \bar{\lambda}^S)$, multiple steady states arise under financial integration. The threshold value $\tilde{w}^S$ is defined as the wage rate at the unstable steady state. $\square$

Proof of Proposition 5

Proof. Under autarky, the law of motion for wage in setting **N** is $w_{t+1} = \left(\frac{R}{\rho} w_t\right)^\alpha$, identical as that in our model with $\sigma = 0$; in setting **F**, the law of motion for wage is

$$w_{t+1} = \begin{cases} \left(\frac{R}{\rho} w_t\right)^\alpha, & \text{for } w_t \in (0, \mathbf{m}); \\ \left(\frac{R\mathbf{m}}{\rho}\right)^\alpha, & \text{for } w_t > \mathbf{m}. \end{cases}$$

Thus, the proof follows exactly the proofs for proposition 1 and lemma 1. $\square$

Proof of Lemma 3

Proof. In setting **N**, if $\tau \in (1 - \lambda, 1)$, the borrowing constraints are slack, $\psi_t = \tau \in (1 - \lambda, 1)$ and, according to equation (A.5), $\frac{\partial \ln r_t}{\partial \ln w_t} = \alpha - 1 < 0$; if $\tau \in (0, 1 - \lambda]$, the borrowing constraints are binding, $\psi_t = \tau \in (0, 1 - \lambda)$, and, according to equation (A.5), $\frac{\partial \ln r_t}{\partial \ln w_t} = \alpha - 1 < 0$.

In setting **F**, if $\psi_t \in (1 - \lambda, 1)$, the borrowing constraints are slack and, according to equation (A.6), $\frac{\partial \ln r_t}{\partial \ln w_t} = \alpha - 1 < 0$. If $\psi_t \in (0, 1 - \lambda)$, the borrowing constraints are binding $\psi_t = \frac{w_t}{\mathbf{m}} \in (0, 1 - \lambda)$. Define $\tilde{\psi}_A \equiv \frac{1-\alpha}{2-\alpha}$. Rewrite equation (A.6)

$$\begin{aligned} \ln r_t^i &= (\alpha - 1) \ln w_t^i - \ln(1 - \psi_t) + \ln \lambda R^\alpha \rho^{1-\alpha}. \\ \Rightarrow \frac{\partial \ln r_t^i}{\partial \ln w_t^i} &= (\alpha - 1) + \frac{\psi_t}{1 - \psi_t} \begin{cases} < 0, & \text{iff } \psi_t \in (0, \tilde{\psi}_A); \\ > 0, & \text{iff } \psi_t \in (\tilde{\psi}_A, 1 - \lambda). \end{cases} \end{aligned}$$

□

Proof of Proposition 6

Proof. For notational simplicity, we suppress the country index i . By assumption, $\mathbf{m} > \left(\frac{R}{\rho}\right)^\rho$ and hence, $w_A = \left(\frac{R}{\rho}\right)^\rho < \mathbf{m}$, implying that only a fraction of agents can become entrepreneurs in the autarkic steady state $\tau_A = \frac{w_A}{\mathbf{m}} < 1$.

Steady-State Property under Financial Integration in Setting N

The borrowing constraints are binding, if $Rq_{t+1}^i > r^*$ or equivalently $\psi_t^i < \bar{\psi} \equiv 1 - \lambda$. Combine equations (1)-(3) with $r_t^i = r^*$ to get the law of motion for wage,

$$r^*(m_t - w_t) = \lambda q_{t+1} R m_t \Rightarrow r^* \left[\frac{\rho}{R} w_{t+1}^{\frac{1}{\alpha}} - \tau w_t \right] = \lambda \rho w_{t+1}, \quad (\text{C.22})$$

$$\frac{\partial w_{t+1}}{\partial w_t} = \frac{w_{t+1}}{w_t} \frac{1}{1 + \frac{w_{t+1}^{\frac{1}{\alpha}}}{R \tau w_t}} \quad \text{and} \quad \frac{\partial^2 w_{t+1}}{\partial w_t^2} = -\frac{w_{t+1}^{\frac{1}{\alpha}-1}}{\tau \alpha R} \left(\frac{\partial w_{t+1}}{\partial w_t} \right)^3 < 0. \quad (\text{C.23})$$

Equation (C.22) implies that, for $w_t \rightarrow 0$, the law of motion for wage has a positive vertical intercept at $w_{t+1} = \lambda \rho w_A \left(\frac{\rho}{r^*}\right)^\rho$. Define $\bar{w}_F = \frac{1-\lambda}{\tau} w_A \left(\frac{\rho}{r^*}\right)^{\frac{1}{1-\alpha}}$.

- For $w_t \in (0, \bar{w}_F)$, the borrowing constraints are binding and the law of motion for wage is upward-sloping and concave, according to equations (C.23).
- For $w_t > \bar{w}_F$, aggregate investment are so high that $Rq_{t+1} = r^*$ and the borrowing constraints are slack. The law of motion for wage is flat at $\bar{w}_{t+1} = w_A \left(\frac{\rho}{r^*}\right)^\rho$.

Use equation (C.23) to evaluate $\frac{\partial w_{t+1}}{\partial w_t}$ at any steady state(s),

$$\frac{\partial w_{t+1}}{\partial w_t} \Big|_{w_{t+1}=w_t} = \begin{cases} \frac{1}{1 + \frac{w_t^{\frac{1}{\alpha}}}{R \tau w_t}} \in (0, 1), & \text{if } w_t \in (0, \bar{w}_F]; \\ 0, & \text{if } w_t > \bar{w}_F. \end{cases} \quad (\text{C.24})$$

Since the slope of the law of motion for wage at any steady state is non-negative and less than unity, the steady state under financial integration must be unique and stable. See figure (14).

Steady-State Property under Financial Integration in Setting F

The proof is essentially the same as that for proposition 3. We first analyze the shape of the law of motion for wage and then describe the conditions for multiple steady states. In particular, we explicitly take into account the mass-of-entrepreneurs constraints.

The borrowing constraints are binding, if $Rq_{t+1} > r^*$ or equivalently $\psi_t < \bar{\psi} \equiv 1 - \lambda$. Combine equations (1)-(3) with $r_t = r^*$ to get the law of motion for wage,

$$\psi_t = \frac{w_t}{\mathbf{m}} = 1 - \lambda \frac{q_{t+1}R}{r^*} = 1 - \frac{\lambda\rho}{r^*} \left(\frac{w_A}{w_{t+1}} \right)^{\frac{1}{\rho}}, \quad (\text{C.25})$$

$$\frac{\partial w_{t+1}}{\partial w_t} = \frac{w_{t+1}}{w_t} \left[1 + \frac{\psi_t - (1 - \alpha)}{(1 - \alpha)(1 - \psi_t)} \right] = \frac{w_{t+1}}{w_t} \frac{\psi_t \rho}{1 - \psi_t} > 0, \quad (\text{C.26})$$

$$\frac{\partial^2 w_{t+1}}{\partial (w_t)^2} = \left(\frac{\partial w_{t+1}}{\partial w_t} \right)^2 \frac{1}{\alpha w_{t+1} w_t} > 0. \quad (\text{C.27})$$

According to equation (C.26), for $w_t \rightarrow 0$, $\psi_t \rightarrow 0$ and the law of motion for wage has a positive vertical intercept at $w_{t+1} = \underline{w}_{t+1} \equiv \lambda^\rho w_A \left(\frac{\rho}{r^*} \right)^\rho$, as in setting N. Combine equation (C.25) with (1)-(2) and then compute the mass of entrepreneurs,

$$w_{t+1} = w_A \left[\frac{\lambda\rho}{r^*(1 - \psi_t)} \right]^\rho \Rightarrow \tau_t = \frac{K_{t+1}}{R\mathbf{m}} = \frac{\rho(w_{t+1})^{\frac{1}{\alpha}}}{R\mathbf{m}} = \frac{w_A}{\mathbf{m}} \left[\frac{\lambda\rho}{r^*(1 - \psi_t)} \right]^{\frac{1}{1-\alpha}}. \quad (\text{C.28})$$

The mass of entrepreneurs is constrained by the population size in each generation,

$$\tau_t \leq 1, \Rightarrow \psi_t \leq \psi_1 \equiv 1 - \left(\frac{w_A}{\mathbf{m}} \right)^{1-\alpha} \frac{\lambda\rho}{r^*}. \quad (\text{C.29})$$

In the following, I characterize the shape of the law of motion for wage, taking into account the borrowing constraints and the mass-of-entrepreneurs constraint.

- If $\psi_1 > \bar{\psi}$ or equivalently $\rho \left(\frac{w_A}{\mathbf{m}} \right)^{1-\alpha} < r^*$, let $\bar{w}_F \equiv (1 - \lambda)\mathbf{m}$.
 - For $w_t \in (0, \bar{w}_F)$, $\tau_t < 1$ and the borrowing constraints are binding, $\psi_t \in (0, \bar{\psi})$. The law of motion for wage is convex and defined by equation (C.25).
 - For $w_t > \bar{w}_F$, the borrowing constraints are slack, $Rq_{t+1} = r^*$. The law of motion for wage is flat at $w_{t+1} = w_A \left(\frac{\rho}{r^*} \right)^\rho$.
- If $\psi_1 < \bar{\psi}$ or equivalently $\rho \left(\frac{w_A}{\mathbf{m}} \right)^{1-\alpha} > r^*$, let $\bar{w}_F \equiv \mathbf{m}\psi_1$.
 - For $w_t \in (0, \bar{w}_F)$, $\tau_t < 1$ and the borrowing constraints are binding, $\psi_t \in (0, \bar{\psi})$. The law of motion for wage is convex and defined by equation (C.25).
 - For $w_t > \bar{w}_F$, the borrowing constraints are binding, $\psi_t < \bar{\psi}$. The law of motion for wage is flat at $w_{t+1} = \left(\frac{R\mathbf{m}}{\rho} \right)^\alpha$.

According to equation (C.26), if $\psi_t > 1 - \alpha$ holds at any steady state in the interval of $w_t \in (0, \min\{\psi_1, \bar{\psi}\})$, the slope of the law of motion for wage at the steady state exceeds unity so that multiple steady states arise. Figure 16 shows the laws of motion for wage under financial integration versus under autarky in five cases, while figure 15 shows the parameter configuration of these five cases in the $\{\lambda, \psi_A\}$ space and in the $\{\lambda, Z\}$ space, respectively. In the following, we derive the boundary conditions for these five regions.

Case 1: Consider the upper-right triangle of figure 15, i.e., $\psi_A \in (\bar{\psi}, 1)$. Given $r^* = r_A = \rho$, the law of motion for wage under financial integration is flat at the initial steady state S. Compare the upper-right and the lower-right panels of figure 16. Multiple steady states arise if the convex part of the law of motion for wage is at least tangent with the 45° line at point M, i.e., $w_t = w_{t+1} = w_M < w_A$ and $\frac{\partial w_{t+1}}{\partial w_t} \big|_{w_M} \geq 1$. Rewrite equations (C.25) and (C.26) to get the conditions that define region BC of figure 15

$$\lambda < \alpha \text{ and } \bar{\psi} < \psi_A \leq \min\{\psi_{BC}, 1\}, \text{ where } \psi_{BC} \equiv (1 - \alpha) \left(\frac{\alpha}{\lambda} \right)^{\frac{\alpha}{1-\alpha}},$$

which is identical as conditions (C.15) in our model with $\sigma = 1$ and $\theta \rightarrow 0$.

Case 2: Consider the lower-left triangular of figure 15 where the borrowing constraints are binding at the autarkic steady state with $\psi_A \in (0, \bar{\psi})$ and $r_A = \frac{\lambda \rho}{1 - \psi_A} < \rho$. As shown in the upper-left panel of figure 16, given $r^* = r_A$ under financial integration, case B arises iff $\frac{\partial w_{t+1}}{\partial w_t} \big|_{w_A} \geq 1$. The solution is

$$\lambda < \alpha \text{ and } \hat{\psi}_F < \psi_A \leq \bar{\psi}, \text{ where } \hat{\psi}_F \equiv 1 - \alpha.$$

which is identical as conditions (C.16) in our model with $\sigma = 1$ and $\theta \rightarrow 0$.

Case 3: consider the region with $\psi_A < \min\{1 - \alpha, 1 - \lambda\}$. Since $\frac{\partial w_{t+1}}{\partial w_t} \big|_{w_A} < 1$, the autarkic steady state is stable under financial integration. Compare the upper-middle and the lower-left panel of figure 16. Multiple steady states arise under financial integration, if the kink point of the law of motion for wage is above the 45° line, i.e., $\bar{w}_{t+1} > \bar{w}_F$. There are two subcases.

- Case 3.1: if $\psi_1 > \bar{\psi}$ or equivalently $(1 - \psi_A)\psi_A^{1-\alpha} < \lambda$, the kink point is at $\bar{w}_F = (1 - \lambda)m$ and $\bar{w}_{t+1} = w_A \left(\frac{1 - \psi_A}{\lambda} \right)^\rho$. $\bar{w}_{t+1} \geq \bar{w}_F$ gives

$$\psi_A(1 - \psi_A)^\rho \geq (1 - \lambda)\lambda^\rho, \psi_A < \min\{1 - \alpha, \bar{\psi}\}, \text{ and } (1 - \psi_A)\psi_A^{1-\alpha} < \lambda, \quad (\text{C.30})$$

which is identical as (C.17) in our model with $\sigma = 1$ and $\theta \rightarrow 0$.

- Case 3.2: if $\psi_1 < \bar{\psi}$ or equivalently $(1 - \psi_A)\psi_A^{1-\alpha} > \lambda$, the kink point is at $\bar{w}_t^F = \psi_1 m$ and $\bar{w}_{t+1} = \frac{w_A}{\psi_A}$. $\bar{w}_{t+1} \geq \bar{w}_F$ gives

$$(2 - \psi_A)\psi_A^{1-\alpha} \geq 1, \psi_A < \min\{1 - \alpha, \bar{\psi}\}, \text{ and } (1 - \psi_A)\psi_A^{1-\alpha} > \lambda, \quad (\text{C.31})$$

which is identical as (C.18) in our model with $\sigma = 1$ and $\theta \rightarrow 0$.

Region AB is defined by conditions (C.30) and (C.31).

□